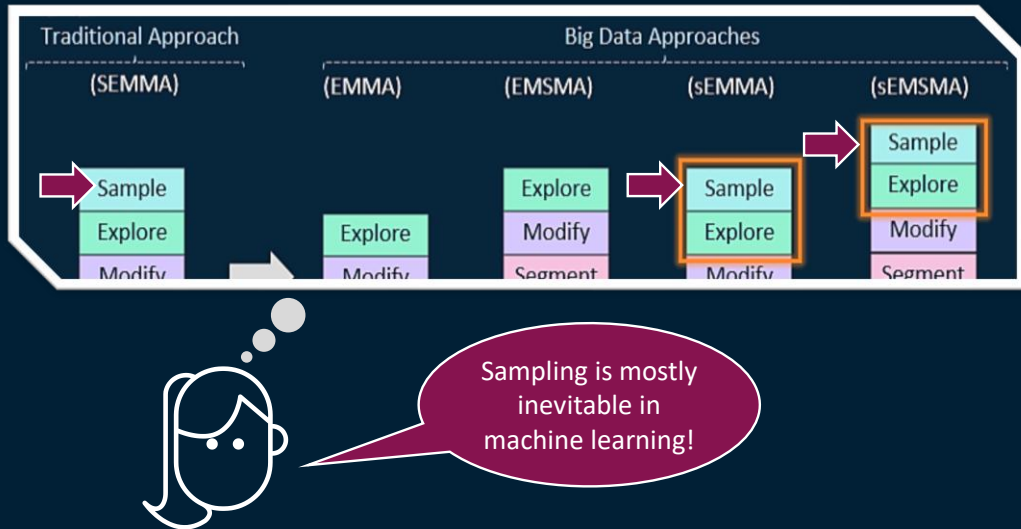


2.1 Introduction to Statistical Analysis



Populations and Samples



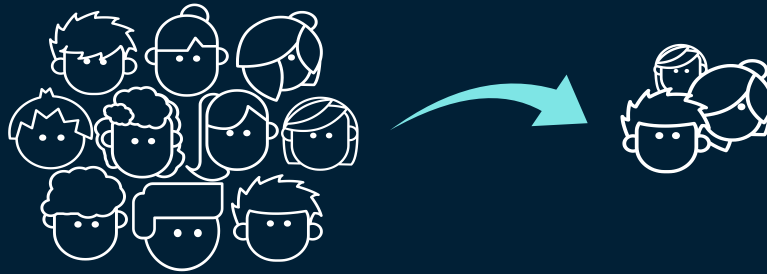
Recall the difference between traditional modeling approaches and big data modeling approaches that we discussed in the previous lesson. Even though SAS Viya can handle big data, it's not entirely correct to say that you don't need to sample when you do machine learning on big data.

Preparing your data for analysis starts with exploring your data. Exploring the data can be time-consuming if processed using the entire data. So, sampling is inevitable.

Populations and Samples

Population – the entire collection of individual members of a group of interest

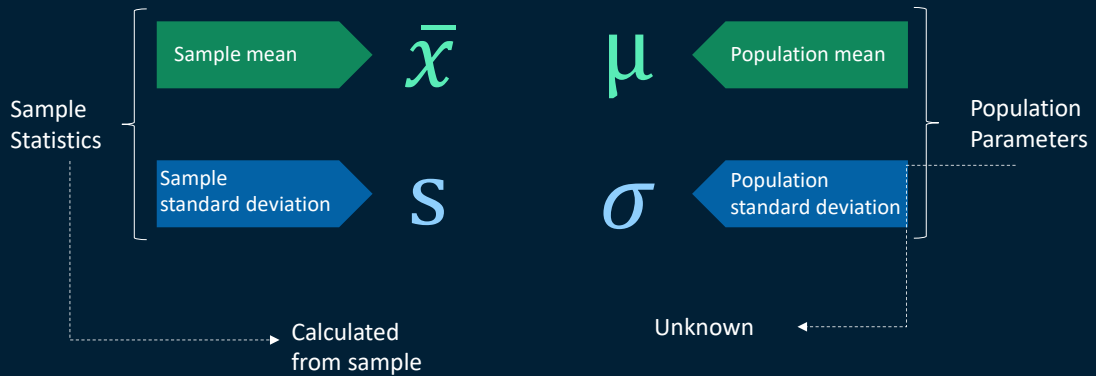
Sample – a subset of a population drawn to enable inferences to the population



A *population* is a collection of *all* objects about which information is desired. Examples of populations are all people with a certain disease, all drivers with a certain level of insurance, or all customers, both current and potential, at a bank. When population sizes are too large to include all possible observations in a statistical test, samples are used instead.

A *sample* is a subset of the population. The sample should be representative of the population, meaning that the sample's characteristics are like the population's characteristics. A thousand bank customers responding to a survey is an example of sample.

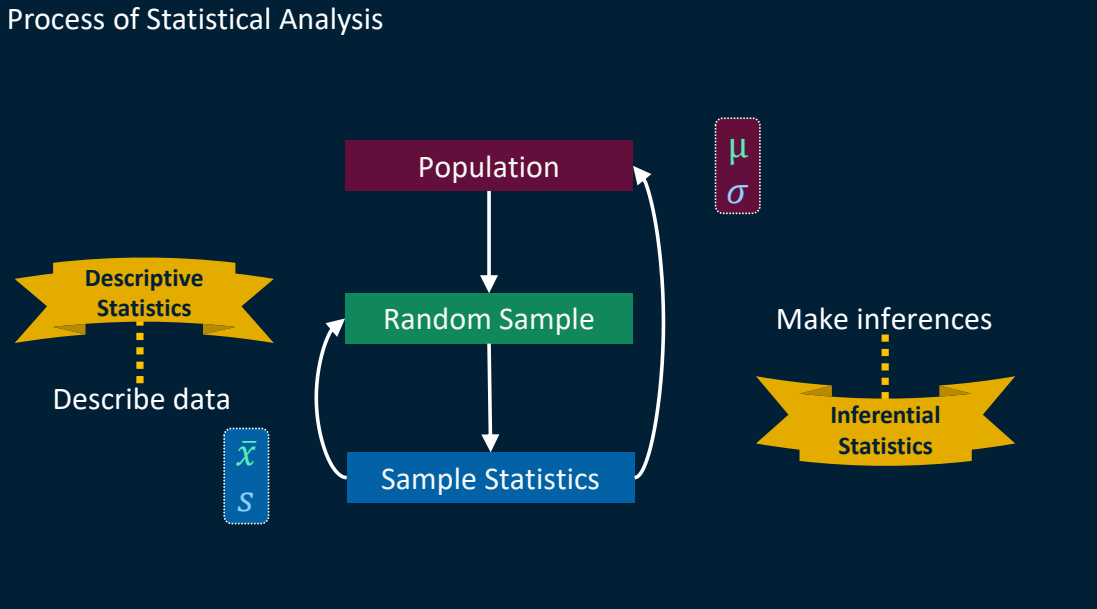
Parameters and Statistics



A descriptive measure of the population is called a *parameter*. *Parameters* are essentially the characteristics of population. Because populations usually cannot be measured in their entirety, parameter values are generally unknown. Parameters are usually denoted by Greek letters. Examples of parameters are population mean (μ), and population standard deviation (σ).

A descriptive measure of a sample is called a *statistic*. Statistics are usually denoted by Roman letters. Examples of statistics are sample mean ($\bar{x}$) and sample standard deviation (s). Statistics are quantities calculated from the values in the sample.

Process of Statistical Analysis



Statistical analysis is the process of collecting and analyzing data in order to draw meaningful conclusions from the raw data. You generally start with identifying the population of interest. And you often want to estimate the value of a parameter or conduct tests about the parameter, for example, μ and σ . However, the calculation of parameters is usually either impossible or infeasible because of the amount of time and money required to reckon the full population. In such cases, you take a random sample of the population, compute sample statistics to describe the sample, such as $\bar{x}$ and s , and use sample information to make inferences about the population by estimating the value of the parameter. The basis for inferential statistics, then, is the ability to make decisions about parameters without enumerating the entire population.

This leads us to describe the two branches of statistics: descriptive statistics and inferential statistics.

If you are using data gathered on a group to describe or reach conclusions about the same group, the statistics are called *descriptive statistics*. Descriptive statistics are used to organize, summarize, and focus on the main characteristics of your data. Summarizing your data in such a manner also makes it more usable. Here, $\bar{x}$ and s are descriptive statistics.

If you gather data from a sample and use the statistics generated to reach conclusions about the population from which the sample was taken, the statistics are *inferential statistics*. Inferential statistics make generalizations or inferences from your data to a larger set of data, based on probability theory. Here, $\bar{x}$ estimates μ , and s estimates σ .

Question: Samples

A sample consists of which of the following?

- a. all units of the population
- b. at least 20% units of the population
- c. at the most 50% units of the population
- d. any fraction of the population

Answer: Samples

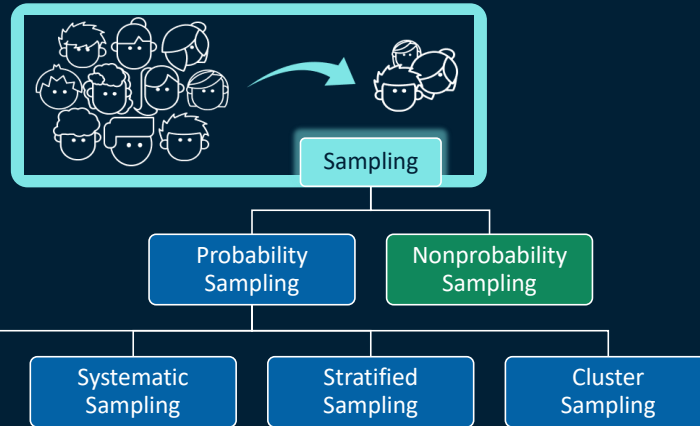
A sample consists of which of the following?

- a. all units of the population
- b. at least 20% units of the population
- c. at the most 50% units of the population
- ☒ d. any fraction of the population

The correct answer is d.

A sample is a smaller part of the whole, that is, a subset of the population. A sample can be any fraction of the population.

Sampling



The process used to extract a sample from a population is called *sampling*. Sampling is broadly classified into *probability* and *non-probability* sampling. In probability sampling, each member of the population has a fixed, known opportunity to belong to the sample, which is not the case with non-probability sampling.

In this course, we focus on four basic sampling techniques: simple random sampling, systematic sampling, stratified sampling, and cluster sampling. These sampling techniques are useful in the machine learning world.

Sampling Methods

Draw a sample of 6 objects out of 18



Consider that you have a population of 18 objects of two types, the colors gray and plum. You want to draw a sample of six objects.

Sampling Methods

Draw a sample of 6 objects out of 18

Simple Random Sampling



The most elementary of the probability sampling is *simple random sampling*. Simple random sampling is a technique in which each element of the population has an equal probability of being selected. All the elements are selected independent of each other. Six objects are randomly selected without considering the color of the objects.

Sampling Methods

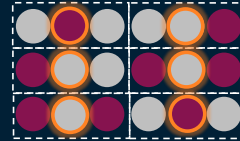
Draw a sample of 6 objects out of 18

Simple Random Sampling



$$\begin{aligned}\text{group size, } k \\ &= N/n \\ &= 18/6 = 3\end{aligned}$$

Systematic Sampling



Systematic sampling involves the selection of objects from an ordered sampling frame. You decide on sample size (n) and then divide frame of N individuals into groups of k individuals where $k = N/n$. It randomly selects a starting point, and then every k th element is selected from the population. For example, here 18 objects are arranged into an ordered sampling frame of six groups, each one of them of size 3. The second object is randomly selected from the first frame and then every third object is selected from the ordered sampling frame.

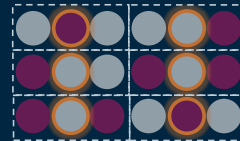
Sampling Methods

Draw a sample of 6 objects out of 18

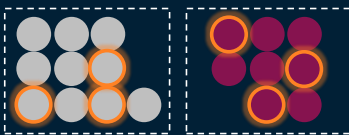
Simple Random Sampling



Systematic Sampling



Stratified Sampling



Stratified sampling is a type of random sampling in which the population is divided into various non-overlapping strata and then elements are randomly selected into the sample from each stratum. Here, the dots are divided into two strata - gray objects and plum objects. Three objects are randomly selected from each stratum.

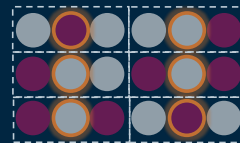
Sampling Methods

Draw a sample of 6 objects out of 18

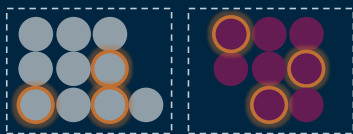
Simple Random Sampling



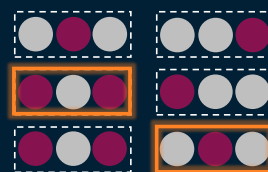
Systematic Sampling



Stratified Sampling



Cluster Sampling



Unlike stratified sampling, *cluster sampling* divides the population into several clusters such that each cluster is representative of the population and then some of the clusters are randomly selected for the sample. Here, the population is divided into six clusters each comprised of gray and plum objects. Then two clusters are randomly selected giving a sample of six objects.

In cluster sampling, the sampling is done on a population of clusters. Therefore, cluster is considered a sampling unit. In stratified sampling, elements within each stratum are sampled. Stratified sampling has homogeneity within strata but heterogeneity between strata whereas cluster sampling has heterogeneity within clusters but homogeneity between clusters.

More about Sampling

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Sampling Methods

In FM radio markets, age of listener is an important determinant of the type of programming used by a station. If a random sample is extracted from each of the three subpopulations of listeners (below 25 years, 25-40 years, and above 40 years), what type of sampling is this?

- a. simple random sampling
- b. stratified sampling
- c. systematic sampling
- d. cluster sampling

Answer: Sampling Methods

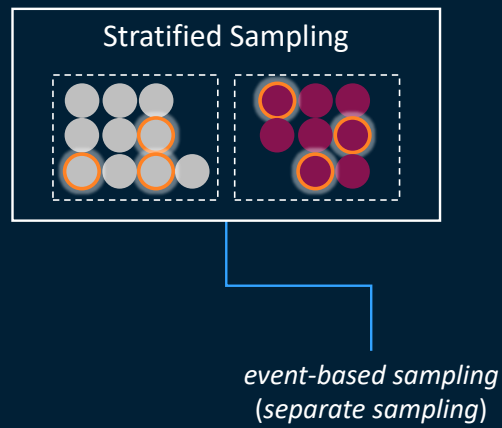
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- a. simple random sampling
- ☒ b. stratified sampling
- c. systematic sampling
- d. cluster sampling

Correct Answer: b

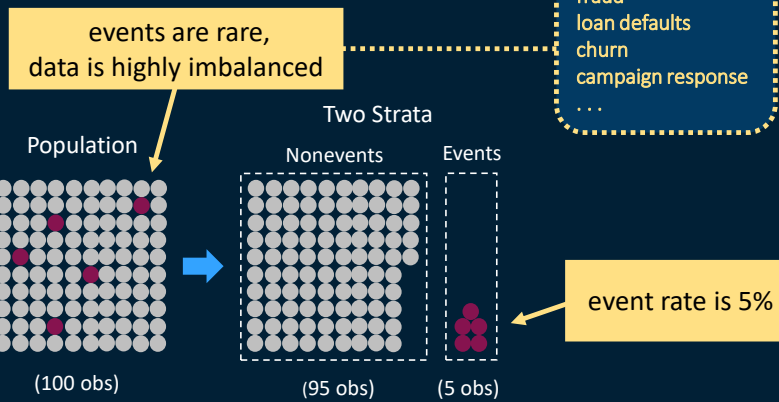
In stratified sampling, the population is divided into nonoverlapping subpopulations called *strata*. The researcher then extracts a random sample from each of the subpopulations.

Event-Based Sampling



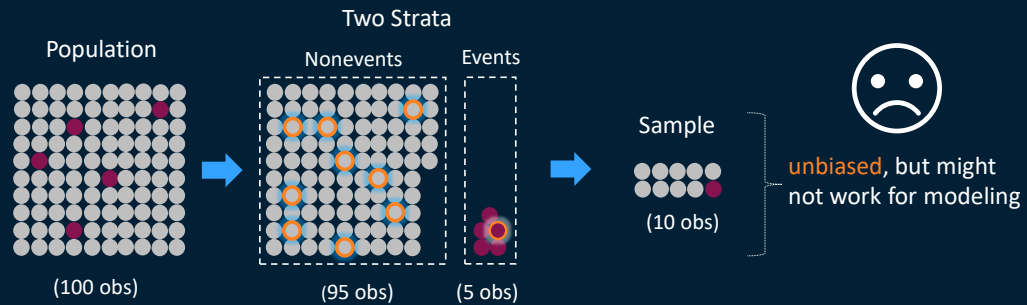
One type of stratified sampling, called *event-based sampling* or *separate sampling*, is extremely important in machine learning predictive models.

Event-Based Sampling



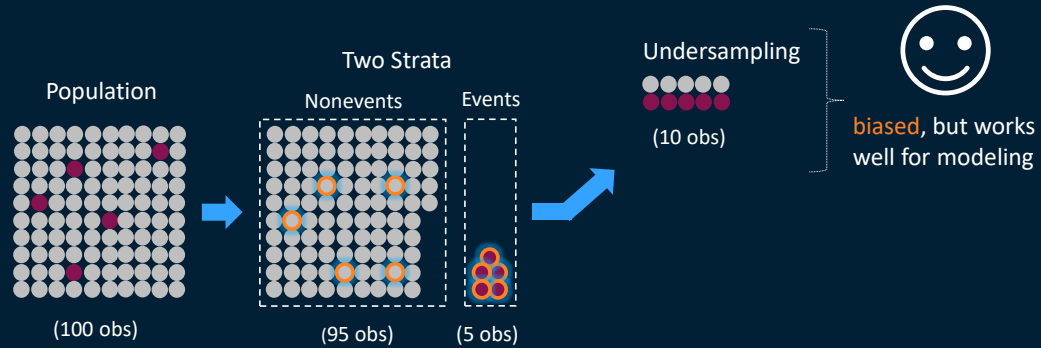
Consider a binary target with categories logically labeled as events (the plum dots) and nonevents (the gray dots). We are interested in building a predictive model with a focus to predict the event. However, events are rare and thus the data is highly imbalanced. Most of the machine learning problems have rare events like predicting fraud, loan defaults, customer churn, and campaign response. For example, the actual event rate is 5%. That means that if there are 100 observations in the population, only five of them are events and 95 are nonevents.

Event-Based Sampling



Event-based samples also known as target-based samples are created by considering the events separately from the nonevents. There are two strata and target-based stratified sampling can be used. A natural practice is to build models from a sample with the same event proportion as there is in the population. For example, to draw a sample of 10 observations, one observation is randomly selected from events and nine observations are randomly selected from nonevents. In this sample, the nonevents far outweigh the events. Although this type of sample would be unbiased, if we train a binary classification model without fixing this problem, the model will be completely biased toward nonevents. For example, a trivial model that predicts every case as a nonevent would be as accurate as 95%. But always predicting nonevents does not help because our primary interest is to build a model to predict the events and not the nonevents.

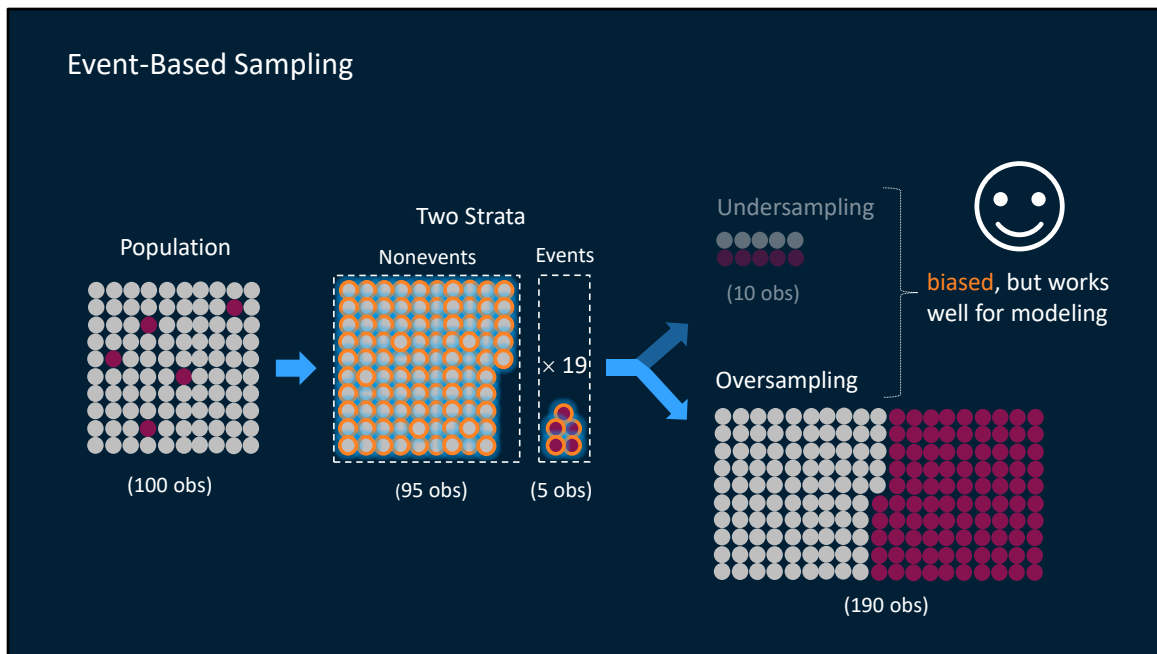
Event-Based Sampling



A common predictive modeling practice is to build models from a sample with an event proportion different from the true population proportion. This is typically done when the ratio of events to nonevents is small. Separate sampling derives its name from the technique used to generate the modeling data, that is, samples are drawn separately based on the target outcome. There are two ways to perform event-based sampling – undersampling and oversampling, both of which have their pros and cons.

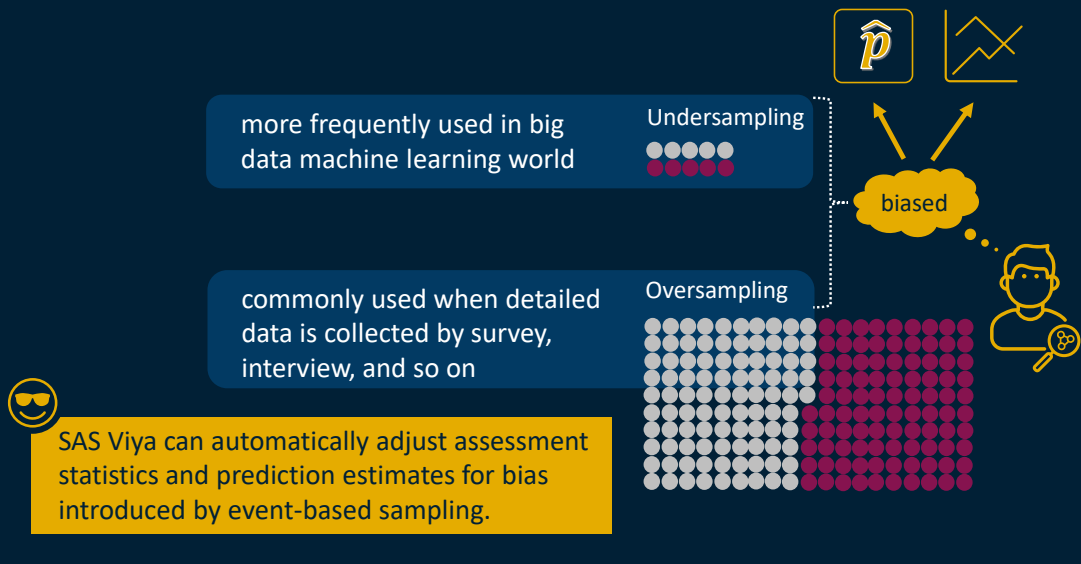
Undersampling is deleting samples from the majority class. In the case of a rare event, usually all events are selected. Then, each event is matched by one or (optimally) more nonevents. For example, to draw a sample of 10 observations, all five events are selected, and an equal number of nonevents are randomly selected from nonevents. In this sample, the events and nonevents are balanced.

Event-Based Sampling



Oversampling, on the other hand, is duplicating the observations of the minority class to obtain a balanced data set. You can select random examples from the minority class with replacement, and supplement the training data with multiple copies of this observation. Hence, it is possible that a single observation can be selected multiple times. Here, a sample of 190 observations is drawn by selecting all the nonevents and including 19 copies of events to make it a balanced sample.

Event-Based Sampling



Oversampling duplicates or creates new synthetic examples in the minority class, whereas undersampling deletes or merges examples in the majority class. Both types of resampling can be effective when used in isolation. Note that some sources describe undersampling as "oversampling" since the event of interest is over-represented in the sample.

Undersampling is generally employed more frequently than oversampling in the big data machine learning world. Overabundance of already collected data is usually encountered in the "big data" scenario, and the reasons to use undersampling are mainly practical and related to resource costs. Specifically, while you need a suitably large sample size to draw valid statistical conclusions, the data must be cleaned before it can be used. Cleansing typically involves significant efforts and time and are therefore costly. So, smaller samples resulting in undersampling are going to be more useful.

Conversely, oversampling is commonly used when the detailed data has yet to be collected by survey, interview, or otherwise. Because of the tremendous cost involved in collecting such information, undersampling is employed much less frequently.

Despite its advantages, event-based sampling (whether oversampling or undersampling) is fundamentally biased and thus introduces some analysis complications.

First, most model fit statistics, especially those related to prediction decisions (like misclassification rate), and some of the assessment plots (like cumulative lift) are closely tied to the outcome proportions in the training samples. If the outcome proportions in the training and validation samples do not match the outcome proportions in the scoring population, model performance can be greatly misestimated.

Second, if the outcome proportions in the training sample and scoring populations do not match, model prediction estimates are biased.

Fortunately, SAS Viya can adjust assessments and prediction estimates to match the scoring population if you specify *prior probabilities*, the scoring population outcome proportions.

Oversampling and Undersampling: Advantages and Disadvantages

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Event-Based Sampling

Both oversampling and undersampling involve introducing bias by having more cases from one class than from another, to compensate for an imbalance representation in the data.

- a. True
- b. False

Answer: Event-Based Sampling

Both oversampling and undersampling involve introducing bias by having more cases from one class than from another, to compensate for an imbalance representation in the data.

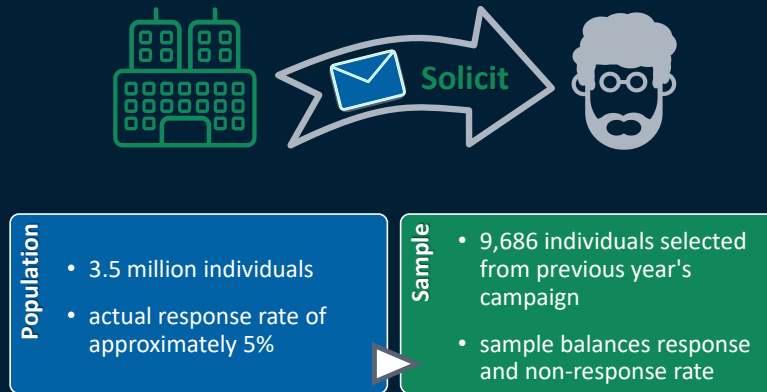
- ☒ a. True
- ☐ b. False

Correct answer: True

Undersampling and oversampling are techniques used to combat the issue of unbalanced classes in a data set. Undersampling induces bias by deleting samples from the majority class, and oversampling induces bias by duplicating samples from the minority class.

Analysis Data

A veterans' organization seeks continued contributions from lapsing donors.



Source: ACM 1998 KDD-Cup competition. Archive at <http://kdd.ics.uci.edu/>.

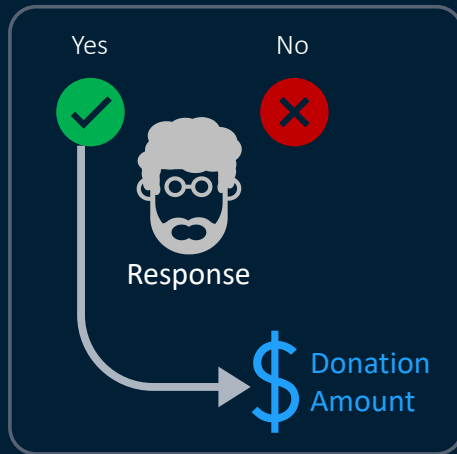
A national veterans' organization wants to better target its solicitations for donation. By soliciting only the most likely donors, the organization can spend less money on solicitation efforts and more money on charitable concerns.

The organization has a mailing database of 3.5 million individuals. These individuals are classified by their response behaviors to previous solicitation efforts. In the original campaign, the response rate was approximately 5%.

The PVA_DONORS table is extracted from a previous year's campaign and is a sample containing 9,686 observations. The data were sampled using event-based sampling so that there are an equal number of responders and non-responders.

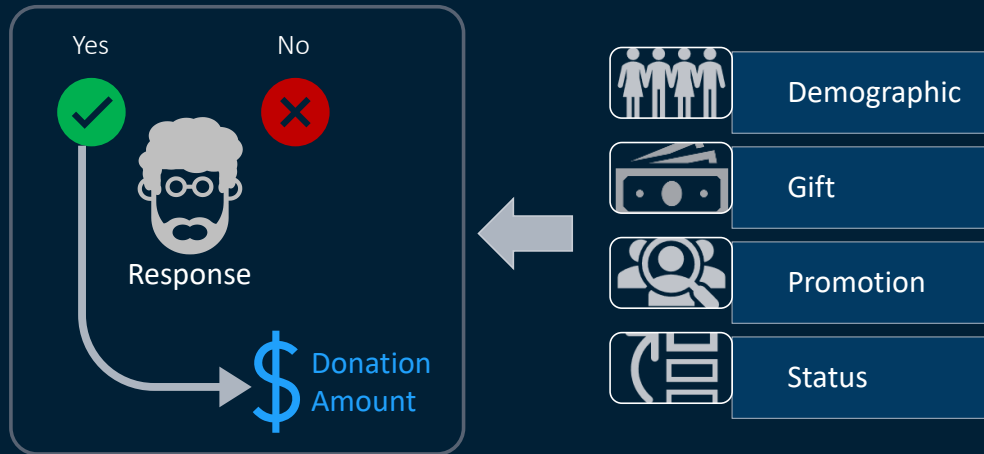
For details about the data, see the Data Dictionary in the course overview.

Analysis Data



The PVA_DONORS data contains two target variables. The binary variable **Response** is a class target that has two levels, *Yes* and *No*. The interval target variable **Donation_Amt** is the value that represents the dollar amount that an individual donates and is associated with those who have responded to the campaign.

Analysis Data



Along with target measurements, several other categorical and continuous measurements are available.

The inputs can be divided into four groups:

- The demographic inputs provide demographic characteristics about the donors and their neighborhoods.
- The gift inputs summarize the previous donation history, providing information about the frequency, size, and recency of past donations.
- The promotion inputs describe the frequency of previous solicitations.
- The status variables categorize individuals into groups, based on recency and frequency measures.

Analysis Goals

Following are the analysis goals of lessons 2, 3, 4, and 5:

2

- Describe the data. Explore for relationships, trends, and anomalies.
- Test whether the mean donation amount is \$100.

→ descriptive statistics

→ inferential statistics

3

- Determine whether demographic, gift, promotion, and status inputs can help explain donation amounts.

→ explanatory modeling
using a continuous target

4

- Use lapsing-donor responses from an earlier campaign to predict future lapsing-donor responses.

→ predictive modeling using
a binary target

5

- Perform some of the data-preprocessing tasks and run a machine learning algorithm.

→ data preprocessing and
machine learning

Let's take a moment to look at the analysis goals for the course.

In Lesson 2, we describe the data and explore for anticipated relationships, unanticipated trends, and anomalies in order to gain understanding and ideas. We then test whether the mean donation amount is \$100.

In Lesson 3, we determine whether demographic, gift, promotion, and status inputs can help explain donation amounts from an earlier campaign.

In Lesson 4, we use lapsing-donor responses from an earlier campaign to predict future lapsing-donor responses.

Finally, in Lesson 5, we run a machine learning algorithm with some of the data-preprocessing tasks.

Demo: Listing Observations

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



2.2 Descriptive Statistics



Describing Your Data

1

Select a random sample of the data.

5 240
68 4021
149 509

2

Use descriptive statistics to better understand your data.

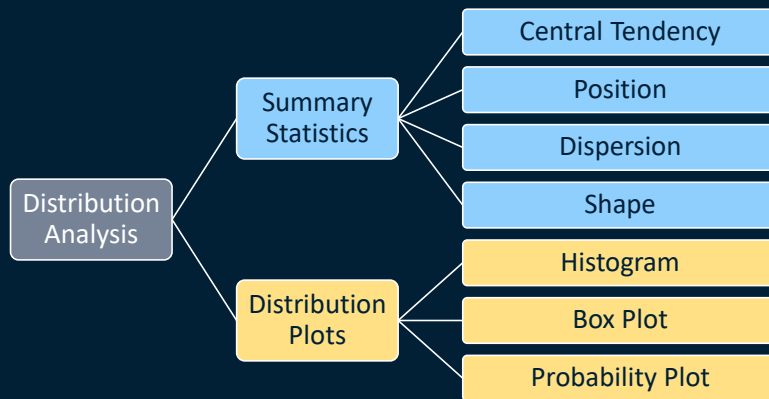


Goals When Describing Data

- ✓ screen for unusual data values
- ✓ inspect the spread and shape
- ✓ characterize central tendency
- ✓ draw preliminary conclusions

After you select a random sample of the data, you can start exploring the data. Although you want to draw conclusions about your population, you first want to describe your data before you use inferential statistics. Descriptive statistics help you to better understand your data by describing and summarizing its basic features. The goals when you're describing data are to screen for unusual data values, inspect the spread and shape of your data, characterize the central tendency, and draw preliminary conclusions about your data.

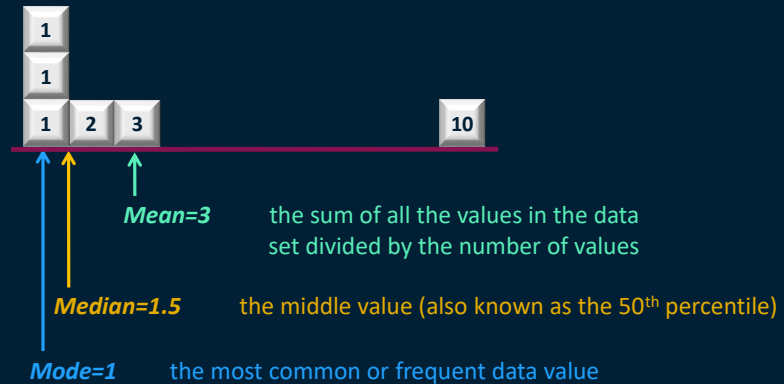
Describing Your Data



Distribution analysis typically includes examining the distribution of values of a continuous variable, the range of possible data values, whether the data values accumulate in the middle of the distribution or at one end, and the shape of the distribution. You use summary statistics and distribution plots for distribution analysis. Summary statistics are numerical statistical measures to describe the data, and distribution plots are graphical displays of data's distribution. Graphical techniques, such as histograms, box plots, and probability plots, enable you to make some general observations about the shape and spread of the data. A more complete understanding of the data can be attained by summarizing the data using statistics. This section presents such statistical measures, including measures of central tendency, measures of position, measures of dispersion, measures of shape, and picturing your distribution.

Measures of Central Tendency

- Mean
- Median
- Mode



Descriptive statistics that locate the center of your data are called *measures of central tendency*. The most common measure of central tendency is the sample mean. Other measures include median and mode. Consider a simple data set with six observations

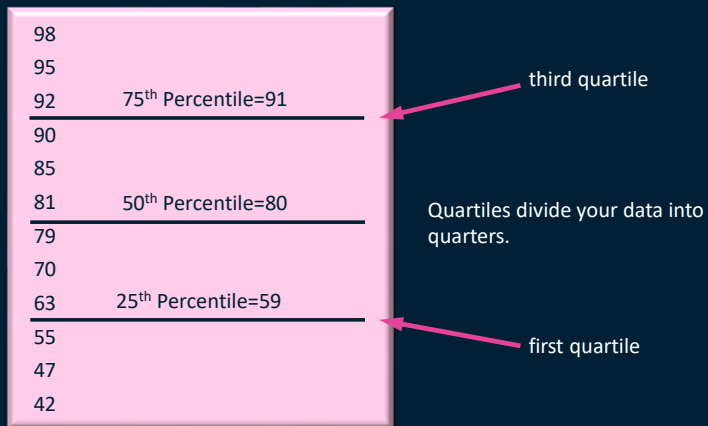
The *mean* is the arithmetic balancing point of your data. It is most appropriate for variables measured on an interval or ratio scale with an approximately symmetrical distribution. Mean is affected by each value in the data set, including the extreme values or outliers. A more robust measure of central tendency that is not affected by extreme values is median.

The *median* is the data point in the middle of a sorted sequence. It is appropriate for either rank scores (variables measured on an ordinal scale) or variables measured on an interval or ratio scale with a skewed distribution. Another measure of central tendency that is not affected by extreme values is mode.

The *mode* is the data point that occurs most frequently. It is most appropriate for variables measured on a nominal scale. There might be several modes in a distribution or there might not be any mode in a distribution.

Measures of Position

- Quartiles
- Deciles
- Percentiles



Measures of position provide you a way to see where a certain data point or value falls in a distribution. A measure can tell us whether a value is about the average or whether it's unusually high or low. Like the median divides the data in two equal halves, quartiles divide the data in four equal parts, deciles divide the data in ten equal parts, and percentiles divide the data in hundred equal parts. *Percentiles* locate a position in your data larger than a given proportion of data values. Commonly reported percentile values are the 25th percentile (also called the first quartile), the 50th percentile (also called the median), and the 75th percentile (also called the third quartile).

Question: Central Tendency

You have the data arranged in ascending order. Which one of the following measures can be affected if the smallest and largest observations are removed?

- a. mean
- b. median
- c. 50th percentile
- d. none of the above

Answer: Central Tendency

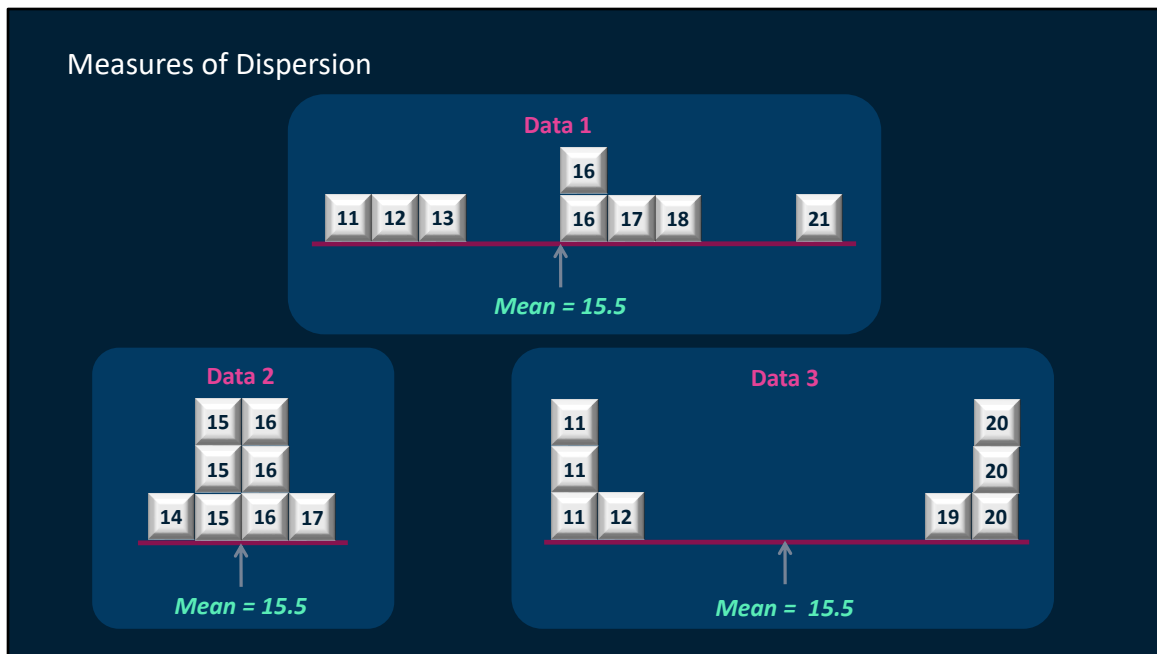
You have the data arranged in ascending order. Which one of the following measures can be affected if the smallest and largest observations are removed?

- ☒ a. mean
- b. median
- c. 50th percentile
- d. none of the above

Mean is affected by every value in the data including largest and smallest observations.

On the other hand, *median* also called as 50th percentile is less affected by extreme values. Because it is the value that divides the data into two equal halves when arranged in order of magnitude.

Measures of Dispersion

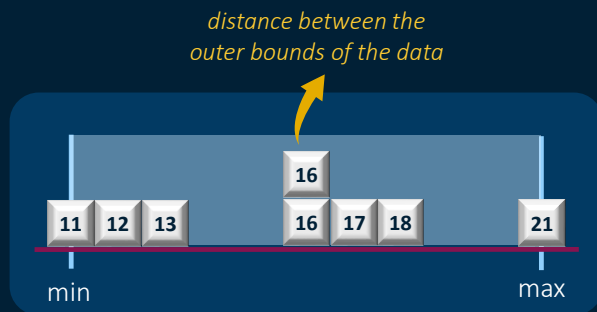


Simply locating the center of the data might not be sufficient. Measures of central tendency yield information only about the center, or middle part, of a group of numbers. Looking at the three data examples, all of them have the same mean, which is 15.5, but it is evident that they have different variability. Using measures of variability in conjunction with measures of central tendency makes possible a more complete numerical description of the data.

Measures of Dispersion

- Range

$$\text{Range} = x_{\max} - x_{\min}$$



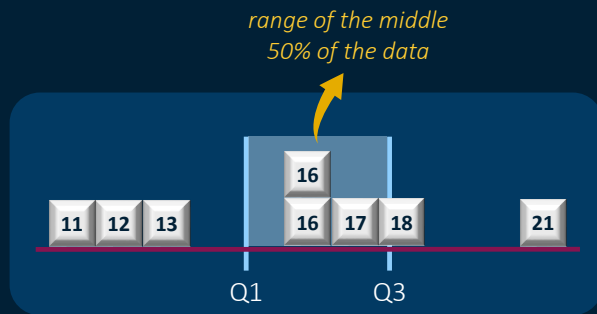
Measures of dispersion enable you to characterize the variability, or spread, of the data.

Range is the simplest measure of dispersion characterized as the difference between the maximum and minimum data values. It is a crude measure of variability, describing the distance between the outer bounds of the data set. A disadvantage of the range is that, because it is computed with the values that are on the extremes of the data, it is affected by extreme values, and its application as a measure of variability is limited.

Measures of Dispersion

- Inter-Quartile Range

$$IQR = Q_3 - Q_1$$



Another measure of variability is the *interquartile range*. The interquartile range is the range of values between the first and third quartile. Essentially, it is the range of the middle 50% of the data and is determined by computing the value of $Q_3 - Q_1$. The interquartile range is especially useful in situations where you are more interested in values toward the middle and less interested in extremes.

Measures of Dispersion

- Mean Absolute Deviation

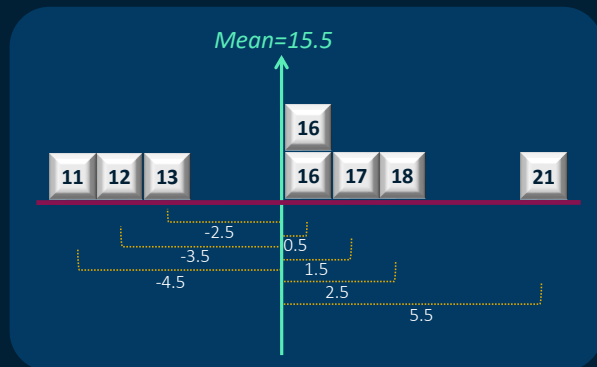
$$MAD = \frac{\sum |x_i - \bar{x}|}{n - 1}$$

- Variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

- Standard Deviation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$



Three other measures of variability are the variance, the standard deviation, and the mean absolute deviation. They are obtained by subtracting the mean from each value of data and are, therefore, presented together. These measures of dispersion essentially yield the deviation from the mean. Geometrically the negative deviations represent values that are below (to the left of) the mean and positive deviations represent values that are above (to the right of) the mean.

The *mean absolute deviation (MAD)* is the average of the absolute values of the deviations around the mean for a set of numbers. Because absolute values are not conducive to easy manipulation, an alternative mechanism for overcoming the zero-sum property of deviations from the mean is devised. This approach utilizes the square of the deviations from the mean. The result is the variance, an important measure of variability. The *variance* is the average of the squared deviations about the arithmetic mean for a set of numbers. The sample variance is denoted by s-square.

Statistics measured in squared units are problematic to interpret. The standard deviation is a more popular measure of variability. The *standard deviation* is the square root of the variance. The population standard deviation is denoted by s . One feature of the standard deviation that distinguishes it from a variance is that the standard deviation is expressed in the same units as the mean and the raw data, whereas the variance is expressed in those units squared.

Measures of Dispersion

Multivariate data
in machine
learning might
require rescaling!

Data 1



Range = 10
Standard Deviation = 3.33

All 3 have
Mean=15.5

Data 2



Range = 3
Standard Deviation = 0.92

Data 3



Range = 9
Standard Deviation = 4.57

In the previous example, the three data sets have the same mean but different measures of variability. Their respective range and standard deviation values are shown.

You encounter multivariate data in machine learning applications in which you have multiple features spanning varying degrees of magnitude, range, and units. A significant issue is that the range of the variables can differ a lot. Using the original scale might put more weight on the variables with a large range. Thus, rescaling might be required to have the equal range and/or variance. Feature scaling is discussed in Lesson 5.

Question: Dispersion

Standard deviation and variance essentially yield the deviation from which of the following?

- a. maximum value
- b. median value
- c. mode value
- d. mean value

Answer: Dispersion

Standard deviation and variance essentially yield the deviation from which of the following?

- a. maximum value
- b. median value
- c. mode value
- ☒ d. mean value

Correct answer: d

The variance and the standard deviation essentially yield the deviation from the mean. The *variance* is the average of the squared deviations about the arithmetic mean for a set of numbers. The *standard deviation* is the square root of the variance.

Question: Summary Statistics

Poor Jose has his head in an oven and his feet in a freezer!

You are a statistician examining the temperature dispersion in different parts of Jose's body. Which of the following statements is reasonable?

- a. On an average, he must feel just fine.
- b. Locating the center of the (temperature) data is sufficient.
- c. Characterizing the variability or spread of the (temperature) data is necessary.
- d. He is a 'hot head' who has 'cold feet'!

Answer: Summary Statistics

Poor Jose has his head in an oven and his feet in a freezer!

You are a statistician examining the temperature dispersion in different parts of Jose's body. Which of the following statements is reasonable?

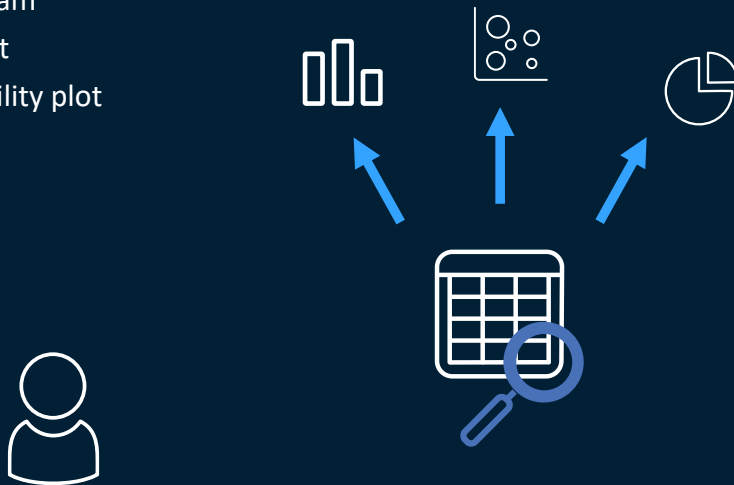
- a. On an average, he must feel just fine.
- b. Locating the center of the (temperature) data is sufficient.
- ☒ c. Characterizing the variability or spread of the (temperature) data is necessary.
- d. He is a 'hot head' who has 'cold feet'!

Correct Answer: c

This is a popular joke about the statistical "average" involving the statistician who put his head in an oven and his feet in a freezer, saying, "On average, I feel fine." Measures of central tendency yield information only about the center, or middle part, of a group of numbers and thus simply locating the center of the data might not be sufficient. Using measures of dispersion in conjunction with measures of central tendency makes possible a more complete numerical description of the data.

Visualizing Distributions

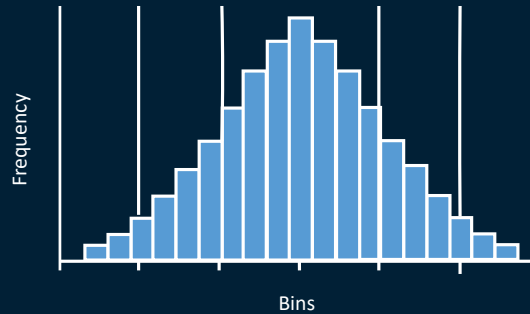
- Histogram
- Box plot
- Probability plot



Another way to understand your data is to display it using graphs and plots. There are several ways to visualize the distribution of your data.

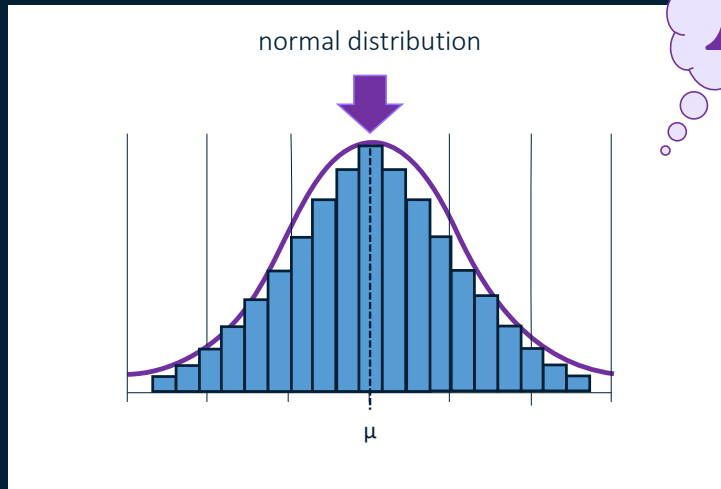
Histograms

visual representation of the
frequency distribution of data



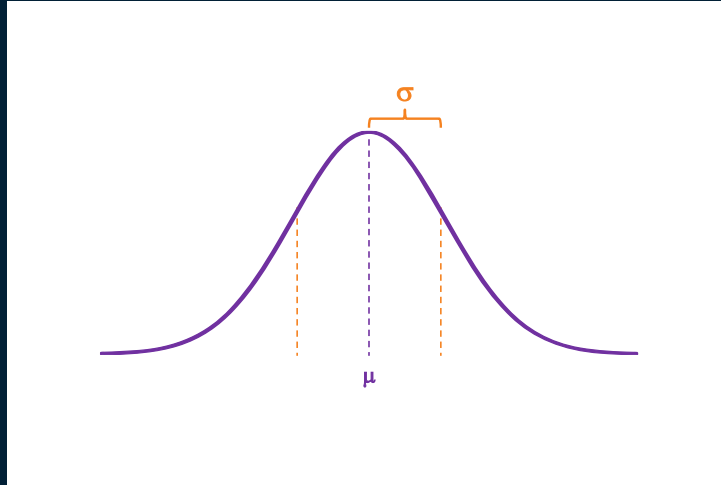
One of the most effective techniques is to plot a histogram. A histogram is a visual representation of the frequency distribution of your data. The frequencies are represented by vertical bars. Each bar in the histogram represents a group of values, sometimes referred to as a bin. The heights of the bars represent the frequency or count of values in the bins.

Histograms



You can easily establish the shape of your data using a histogram. For example, here it shaped like a bell, with values concentrated in the middle. This type of distribution is called a normal distribution. The normal distribution is a common theoretical distribution in statistics.

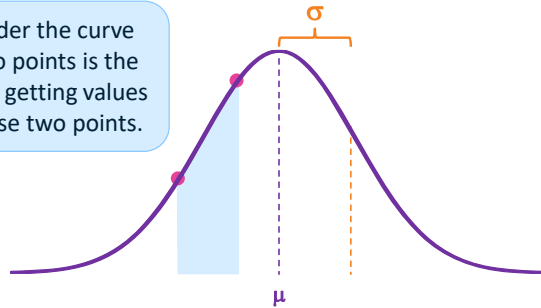
Normal (Gaussian) Distribution



Another name for the normal distribution is the Gaussian distribution. The Gaussian distribution is bell shaped, symmetric, and defined by two parameters, μ (the mean) and σ (the standard deviation). The mean locates the midpoint of the distribution, or the peak of the bell, and the standard deviation describes the variability, or spread of the distribution.

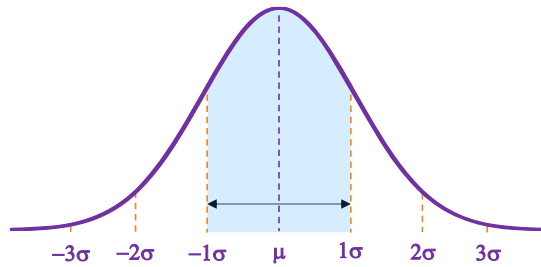
Normal (Gaussian) Distribution

The area under the curve between two points is the probability of getting values between those two points.



The area under the curve between two points is the probability of getting values between those two points.

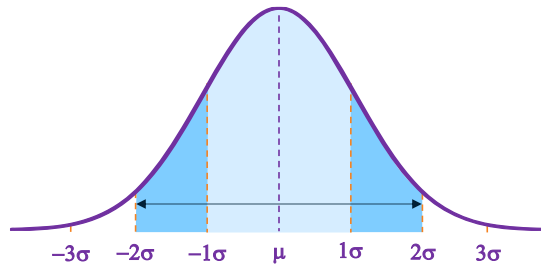
Normal (Gaussian) Distribution



Approximately **68%** of the normal distribution lies within **one** standard deviation of the mean.

Some well-known probabilities are associated with the mean and standard deviation of the distribution. For example, approximately 68% of the total area under normal distribution lies within ± 1 standard deviation of the mean.

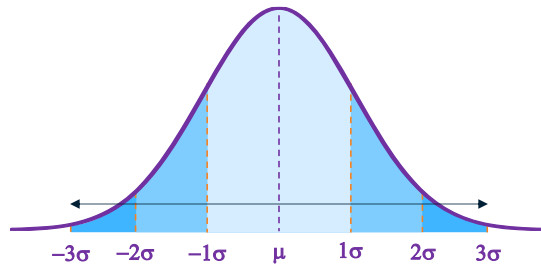
Normal (Gaussian) Distribution



Approximately **95%** of the normal distribution lies within **two** standard deviations of the mean.

Approximately 95% lies within ± 2 standard deviations of the mean.

Normal (Gaussian) Distribution

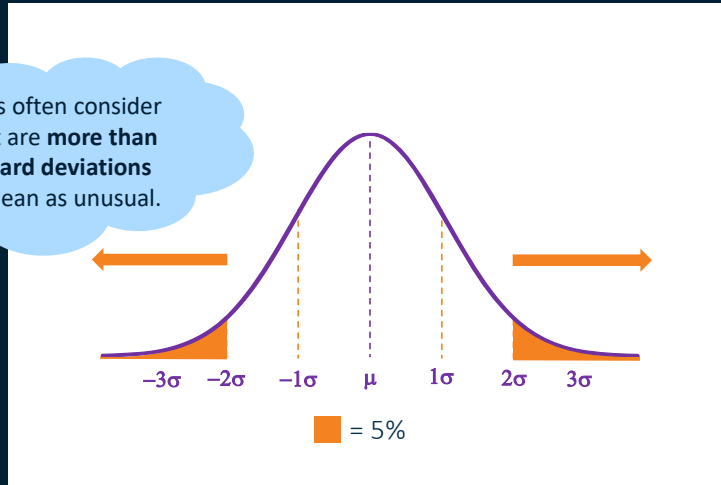


Approximately **99.7%** of the normal distribution lies within **three** standard deviations of the mean.

Approximately 99.7% lies within ± 3 standard deviations.

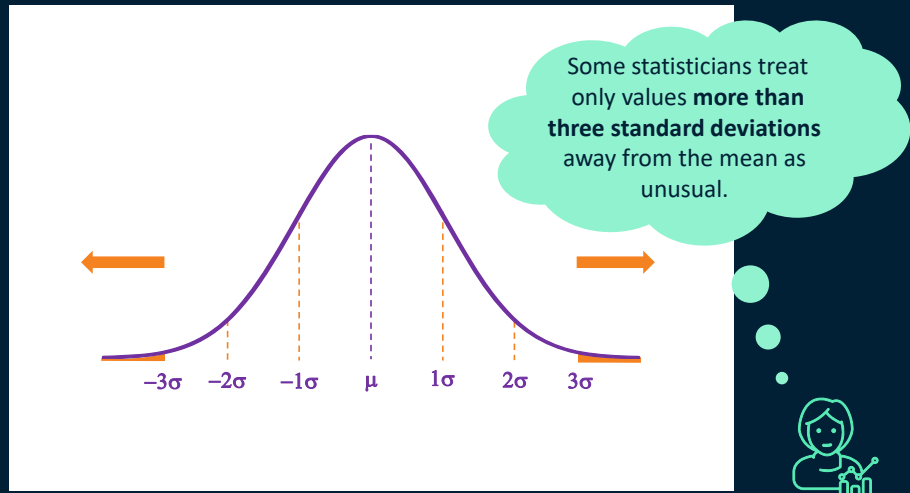
Normal (Gaussian) Distribution

Statisticians often consider values that are **more than two standard deviations** from the mean as unusual.



Statisticians often consider values that are more than ± 2 standard deviations from the mean as unusual. And now you can see why. Only about 5% of all values are that far away from the mean.

Normal (Gaussian) Distribution



Depending on the context, some statisticians treat only values more than ± 3 standard deviations away from the mean as unusual.

Usefulness of Normal Distribution in Machine Learning

data preparation for a machine learning model

Specify a number of standard deviations from the mean to use in deriving lower and upper limits ...

Small | | | | | Large



...to filter and replace extreme observations.

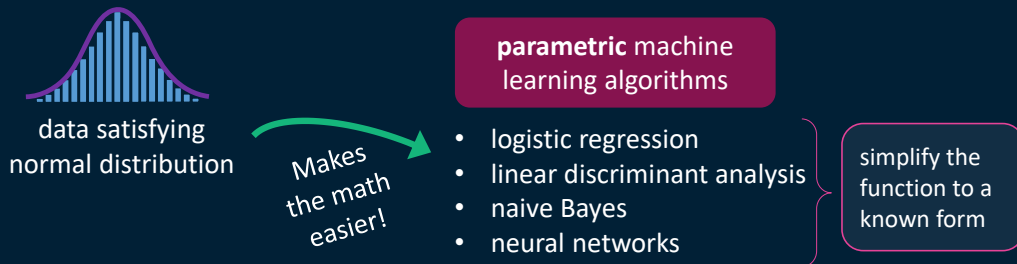


Because the normal distribution has many useful mathematical properties, statistical procedures for data based on a random sample often assume the normal distribution.

Normal distribution assumption is useful in many machine learning tasks including data preparation. You specify the number of standard deviations from the mean to be used in deriving the lower and upper limits for filtering and replacing extreme observations.

Usefulness of Normal Distribution in Machine Learning

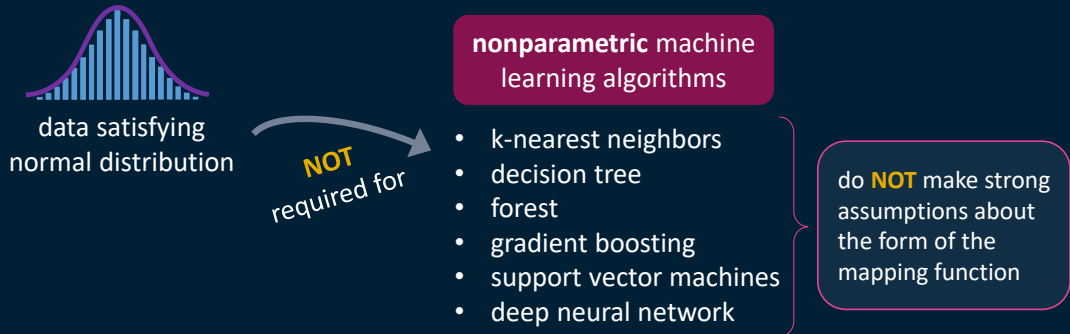
Model Building



Data satisfying normal distribution is beneficial for model building. It makes math easier for models like logistic regression, linear discriminant analysis, naive Bayes, neural networks, and so on. Such algorithms simplify the function to a known form and are called parametric machine learning algorithms. By applying transformation techniques, we can convert the data into a normal distribution.

Usefulness of Normal Distribution in Machine Learning

Model Building



Data satisfying normal distribution is not required with nonparametric models like k-nearest neighbors, decision trees, forest, gradient boosting, support vector machines, deep neural networks etc. Algorithms that do not make strong assumptions about the form of the mapping function are called nonparametric machine learning algorithms.

Question: Normal (Gaussian) Distribution

Which of the following are typical characteristics of a normal distribution?
(Select all that apply)

- a. multimodal
- b. symmetrical, bell shaped
- c. mean, median, and mode all coincide
- d. approximately 99.7% of values lie within ± 3 standard deviation from the mean

Answer: Normal (Gaussian) Distribution

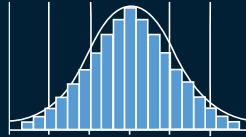
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- ☒ c. mean, median, and mode all coincide
- ☒ d. approximately 99.7% of values lie within ± 3 standard deviation from the mean

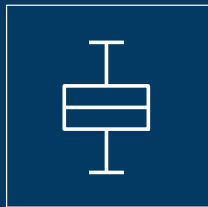
Correct answer: b, c and d.

A normal distribution is unimodal, symmetrical, and bell shaped in which mean, median, and mode are equal. Its empirical rule suggests that approximately 99.7% of values lie within ± 3 standard deviation from the mean.

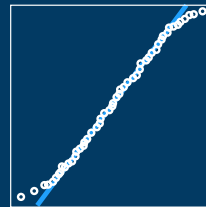
Plots beyond Histograms



Box Plot



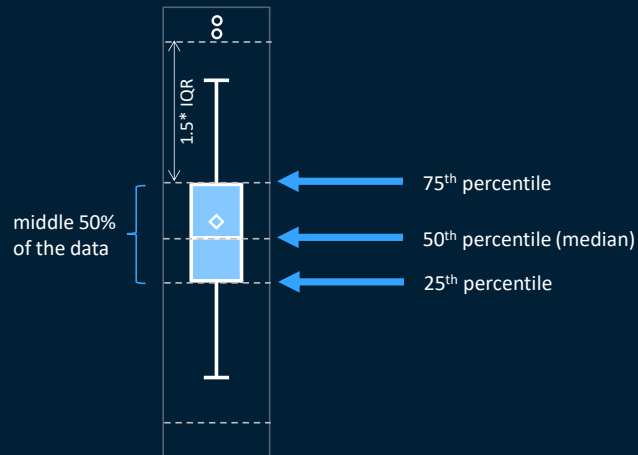
Normal Probability Plot



In addition to plotting your data using a histogram, there are other ways to visualize and assess the distribution of your data. You can use box plots and normal probability plots to compare your data with the normal distribution.

Plots beyond Histograms

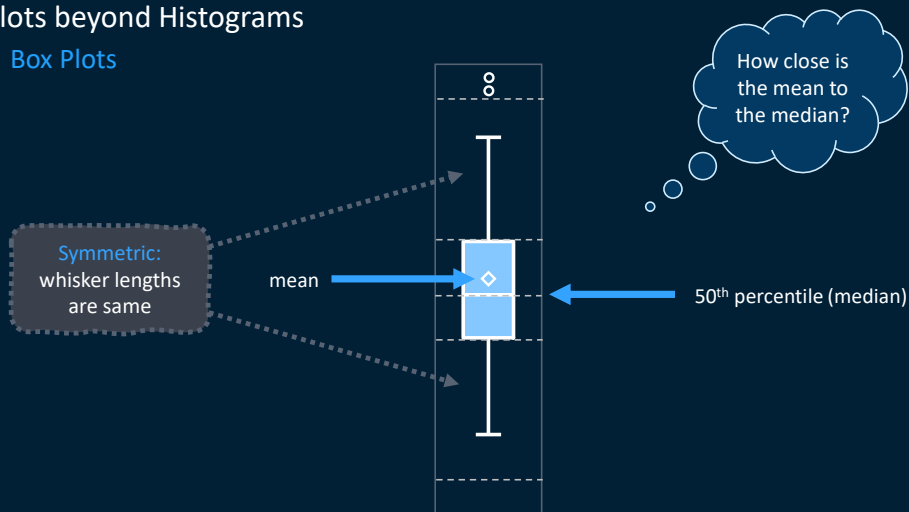
Box Plots



Box plot makes it easy to see how spread out your data is and if there are any outliers. The box represents the middle 50% of your data, which is the interquartile range. The bottom of the box indicates the 25th percentile. The middle line represents the 50th percentile, or the median, and the top line indicates the 75th percentile.

Plots beyond Histograms

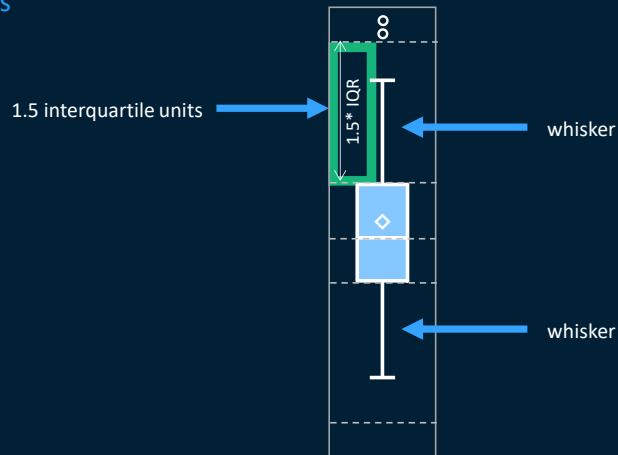
Box Plots



The diamond denotes the mean. Thus, you are able to get a rough impression of the symmetry of your distribution by comparing the mean and median, as well as by assessing the symmetry of the box and whiskers around the median line. The median line is close to the center of the box and the whisker lengths are the same. This box plot shows a data distribution that is approximately symmetric.

Plots beyond Histograms

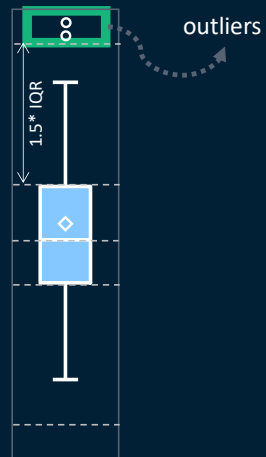
Box Plots



The whiskers extend from the box as far as the data extends, to a maximum length of 1.5 times the interquartile range (IQR) above the 75th percentile and similarly below the 25th percentile on the other side. This is referred to as 1.5 interquartile units.

Plots beyond Histograms

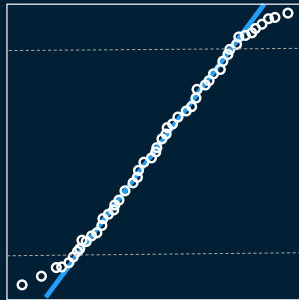
Box Plots



Any data points farther than 1.5 interquartile units away from the box are considered possible outliers and are represented in this plot as circles.

Plots beyond Histograms

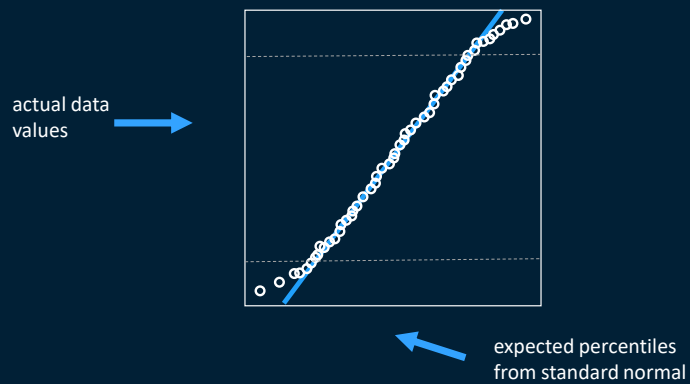
Normal Probability Plots



A normal probability plot is a visual method for determining whether your data comes from a distribution that is approximately normal.

Plots beyond Histograms

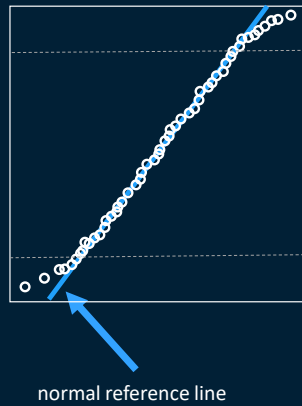
Normal Probability Plots



The vertical axis represents the actual data values, and the horizontal axis displays the expected percentiles from a standard normal distribution.

Plots beyond Histograms

Normal Probability Plots



The normal reference line along the diagonal is where your data would fall if it were perfectly normal. The points hovering around the reference line are the actual data. Here the distribution of data points follows the normal reference line quite closely. Thus, this data distribution looks fairly normal.

Why “1.5” in IQR Method of Outlier Detection?

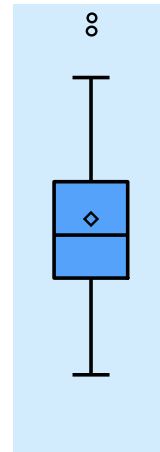
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Question: Visualizing Distributions

In SAS, how is the median indicated in a box plot?

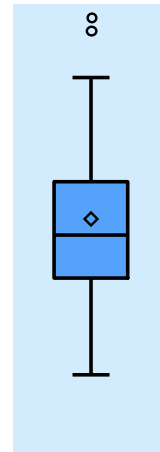
- a. a box
- b. a whisker
- c. a line in the box
- d. a point outside 1.5 IQR



Answer: Visualizing Distributions

In SAS, how is the median indicated in a box plot?

- a. a box
- b. a whisker
- ☒ c. a line in the box
- d. a point outside 1.5 IQR

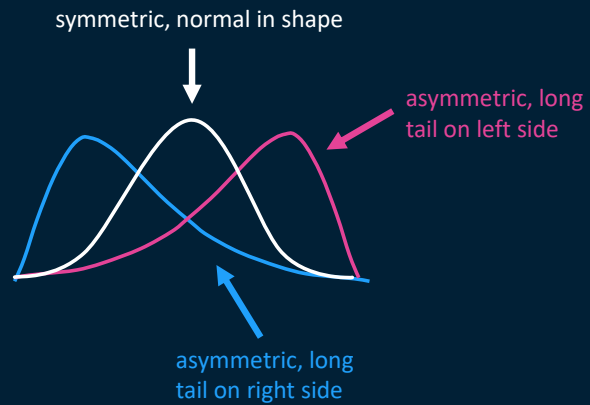


Correct answer: c

The middle line in the box represents the median of the data, whereas the box represents the middle 50% of your data, which is the interquartile range.

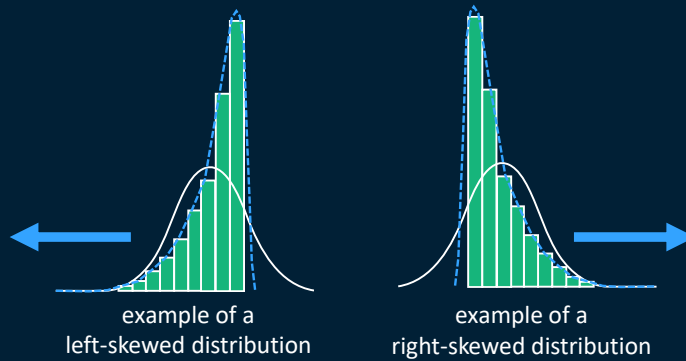
Measures of Shape: Skewness

Skewness: amount of asymmetry of a distribution



Simply ascertaining the dispersion of the data might not be sufficient. Measures of dispersion yield information only about the variability, or spread, of a group of numbers. Looking at the three data distributions, all of them have the equal variability, but it is evident that they have different shapes. Not every distribution is a normal distribution. It is important to determine whether the data is symmetric or asymmetric. When data is asymmetric, the distribution tail on one side is longer than on the other side. Using measures of shapes like skewness help understand the tendency of your data to be more spread out on one side of the mean than on the other. Skewness is a measure of the asymmetry of the distribution. Skewness is when a distribution is asymmetrical or lacks symmetry. A distribution of data in which the right half is a mirror image of the left half is said to be symmetrical. One example of a symmetrical distribution is the normal distribution.

Measures of Shape: Skewness



Here are examples of two skewed distributions.

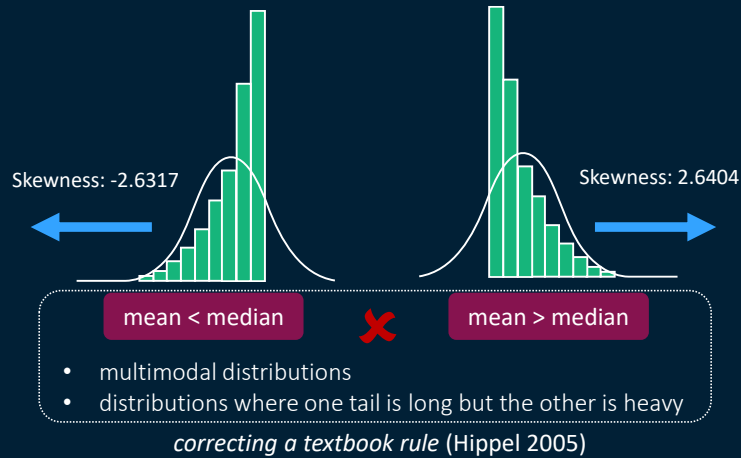
The histogram on the left is an example of a left-skewed distribution, shown by a blue dotted curve. Most values are clustered around the right tail of the distribution, and the left tail of the distribution is longer.

The histogram on the right is an example of a right-skewed distribution, again shown by a blue dotted curve. Most values are clustered around the left tail of the distribution, while the right tail of the distribution is longer.

The white curve on each of these graphs represents the shape of the normal distribution with the mean and variance estimated from the sample data. You can think of the direction of skewness as the direction the data is trailing off to.

Measures of Shape: Skewness

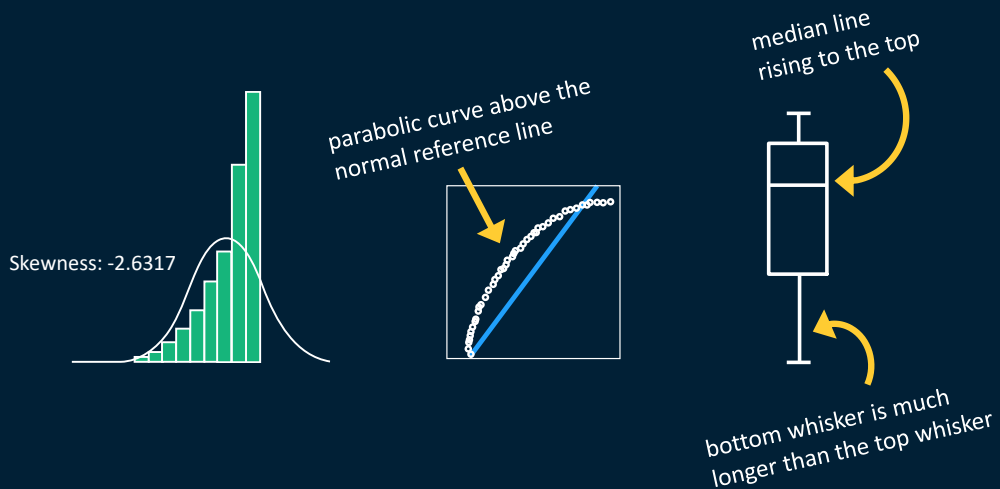
The closer the skewness statistic is to 0, the more normal or symmetric the data.



The closer the skewness statistic is to 0, the more normal or symmetric the data. When the statistic is negative, the data is left-skewed. When the statistic is positive, the data is right-skewed.

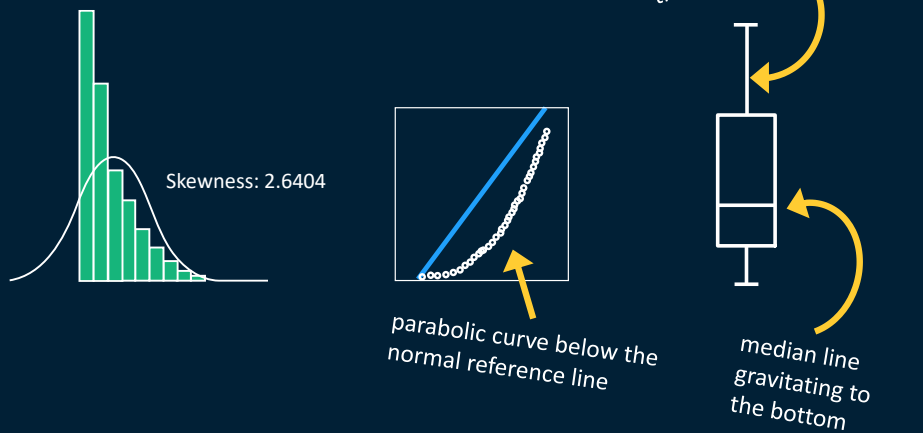
A left-skewed distribution tells us that the mean is less than the median. A right-skewed distribution tells us that the mean is greater than the median. This rule-of-thumb holds for many familiar continuous distributions, such as the exponential, lognormal, and chi-square distributions. However, this relationship does not hold in general with multimodal distributions or in distributions where one tail is long but the other is heavy.

Measures of Shape: Skewness



You can also gauge the shape of the distribution by normal probability plots and box plots. The typical normal probability plot of a left-skewed distribution shows a parabolic curve above the normal reference line. In the box plot of a left-skewed distribution, the bottom whisker is much longer than the top whisker and the median line is rising to the top of the box.

Measures of Shape: Skewness



The typical normal probability plot of a right-skewed distribution shows a parabolic curve below the normal reference line. In the box plot of a right-skewed distribution, the top whisker is much longer than the bottom whisker and the median line is gravitating toward the bottom of the box.

Question: Skewness

Which of the following are true in case of a positively skewed distribution?
(Select all that apply)

- a. it's an asymmetric distribution
- b. mean = median = mode
- c. most values are clustered around the left tail of the distribution while the right tail of the distribution is longer
- d. it is also called left-skewed distribution

Answer: Skewness

Which of the following are true in case of a positively skewed distribution?
(Select all that apply)

- ☒ a. it's an asymmetric distribution
- ☐ b. mean = median = mode
- ☒ c. most values are clustered around the left tail of the distribution while the right tail of the distribution is longer
- ☐ d. it is also called left-skewed distribution

Correct answer: a and c.

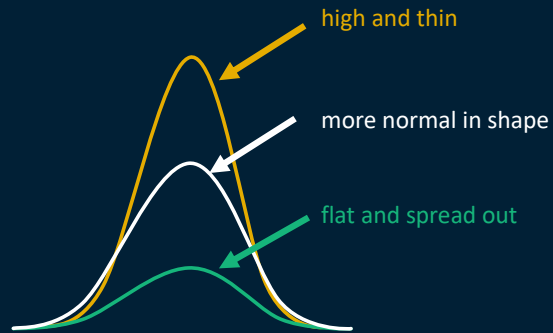
A positively skewed (or right-skewed) distribution is a type of distribution in which most values are clustered around the left tail of the distribution and the right tail of the distribution is longer and the mean is greater than the median thus asymmetric.

Measures of Shape: Kurtosis

Kurtosis: measure of the "peakedness" of the data

Kurtosis: measure of the degree to which a distribution can produce outliers

tail extremity (Westfall 2014)



Again, merely ensuring that the data distribution is symmetrical might not be sufficient. Looking at the three data distributions, they all have the same central tendencies as well as equal variability. Also, all three are symmetrical distributions, but it is evident that they have different *peakedness*. Kurtosis is sometimes said to measure the peakedness of a distribution as opposed to its flatness. A distribution of data that is neither high-and-thin nor flat-and-spread-out is said to be more normal in shape.

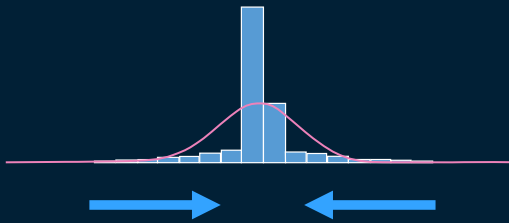
However, Westfall (2014) and other authors have pointed out that this interpretation is incorrect. Westfall makes the following distinction:

Kurtosis tells you virtually nothing about the shape of the peak—its only unambiguous interpretation is in terms of tail extremity, that is, either existing outliers (for the sample kurtosis) or propensity to produce outliers (for the kurtosis of a probability distribution).

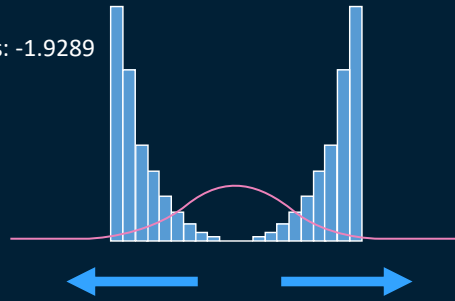
Measures of Shape: Kurtosis

Examples of Excess Kurtosis Distributions

Kurtosis: 6.5557



Kurtosis: -1.9289



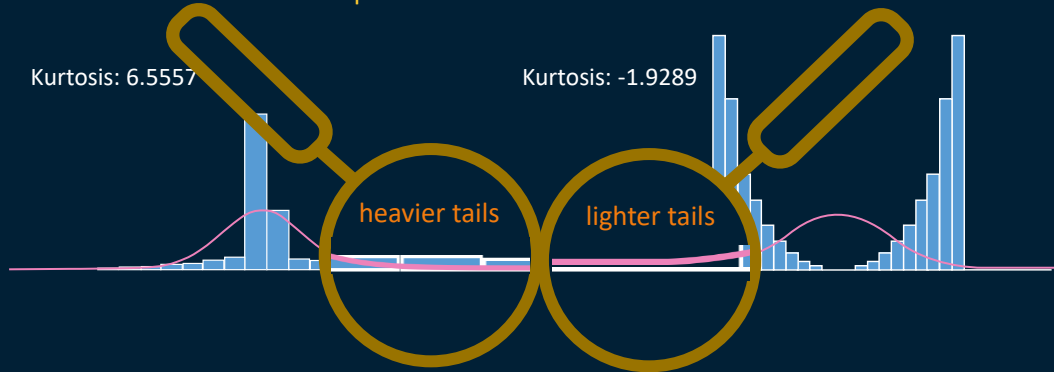
 In SAS, a normal distribution has a zero-kurtosis value.

Here are examples of two excess kurtosis distributions. The graph on the left is an example of a positive-excess kurtosis distribution. The graph on the right is an example of a negative-excess kurtosis distribution. **Kurtosis** measures the tendency of your data to be concentrated toward the center or toward the tails of the distribution. The closer the kurtosis statistic is to 0, the closer the tails of your data resemble the tail thickness of the normal distribution. The pink curve on each of these graphs represents the shape of the normal distribution with the mean and variance estimated from the sample data.

Be aware that normal distribution actually has a kurtosis value of positive 3. SAS has standardized it to zero, so that when you are assessing normality, the values against which to compare the sample skewness and kurtosis are the same.

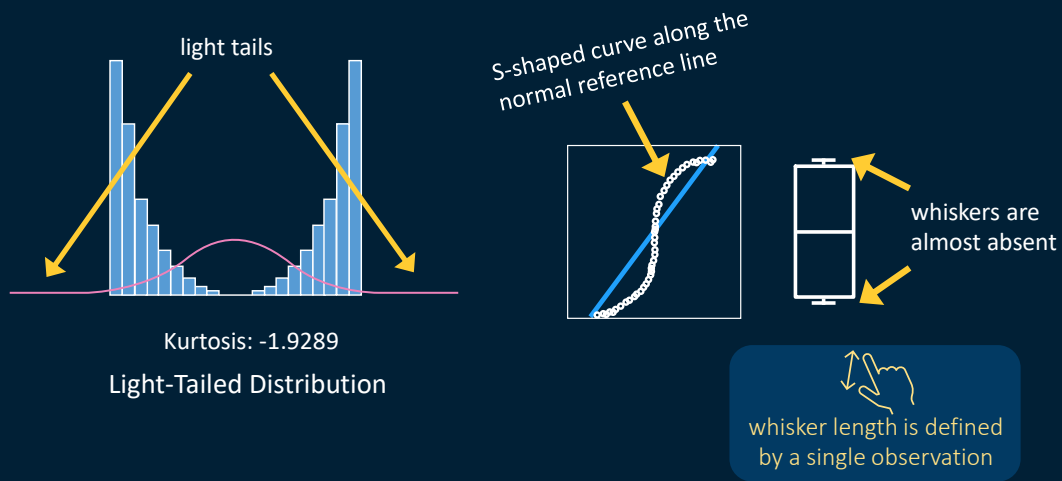
Measures of Shape: Kurtosis

Examples of Excess Kurtosis Distributions



A positive kurtosis statistic means that the data has heavier tails and is more concentrated about the mean than a normal distribution. A negative kurtosis statistic means that the data has lighter tails and is less concentrated about the mean than a normal distribution. If you were to zoom in on the positive kurtosis histogram shown on the left, you'd see that the data extends well beyond two standard deviations of the mean, having more mass in the tails, which are "fatter" or "heavier." Similarly, on the negative kurtosis histogram shown on the right, you'd see less mass in the tails, which are "thinner" or "lighter."

Measures of Shape: Kurtosis



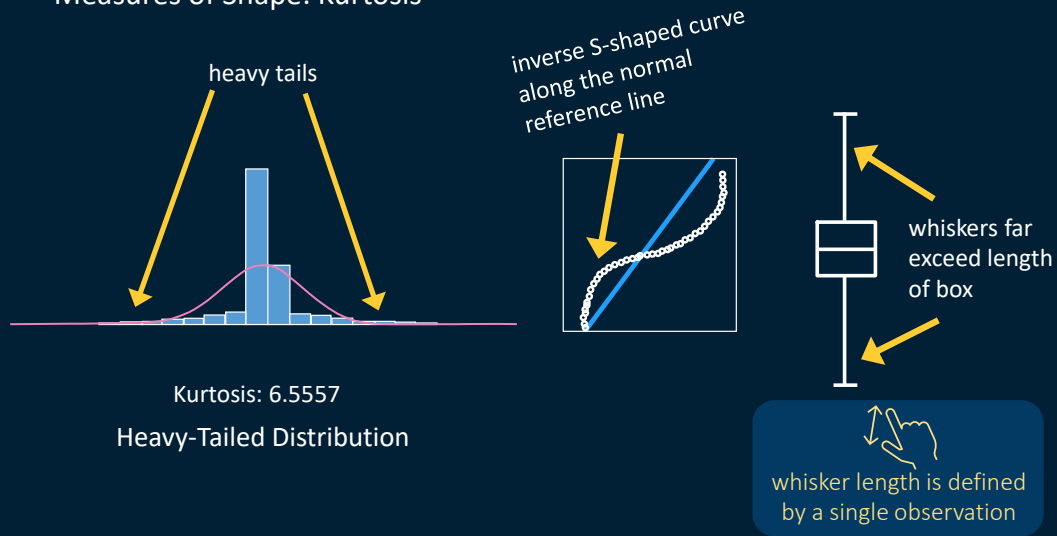
Let's see the shape of the distribution using normal probability plots and box plots.

The histogram shows a light-tailed distribution.

The typical normal probability plot of such a negative-excess kurtosis distribution shows an S-shaped curve along the normal reference line. Remember that the vertical axis is your ordered response values, so the mean of your data is a horizontal line running approximately through the middle of the plot. If you examine your data points paying particular attention to the tails, the data in the tails looks like it is being squashed from the top and the bottom, toward the mean. In other words, the data points are closer to the mean than they should be if they were normal, meaning that the tails of the distribution are light.

The box plot of a negative-excess kurtosis distribution has extremely short tails (actually a U-shaped population, with a dip in the middle rather than a hump). The whiskers are almost absent. However, it is advised not to put too much emphasis on whisker length because the length is defined by a single observation.

Measures of Shape: Kurtosis



The histogram shows a heavy-tailed distribution. The typical normal probability plot of such a positive-excess kurtosis distribution shows an inverse S-shaped curve along the normal reference line. If you examine your data points paying particular attention to the tails, the data in the tails looks like it is being stretched in the vertical direction, away from the mean. In other words, the data points are farther from the mean than they should be if they were normal, meaning the tails of the distribution are heavy. The box plot of a positive-excess kurtosis distribution has long tails. The length of the whiskers far exceeds the length of the box.

Platykurtic and Leptokurtic Distributions

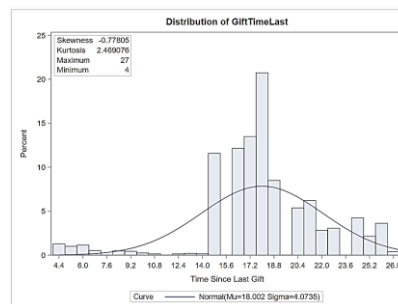
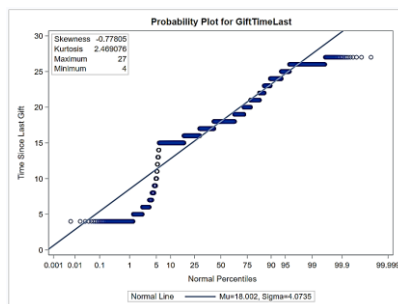
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Question: Normal Distribution

Examine the normal probability plot and histogram of Time Since Last Gift. Do the plots indicate that the data is approximately normally distributed?

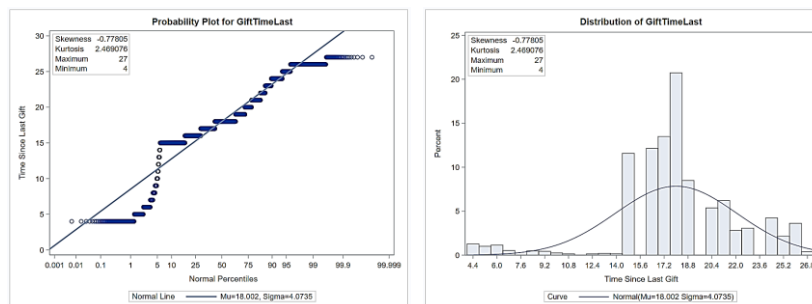
- a. True
- b. False



Answer: Normal Distribution

Examine the normal probability plot and histogram of Time Since Last Gift. Do the plots indicate that the data is approximately normally distributed?

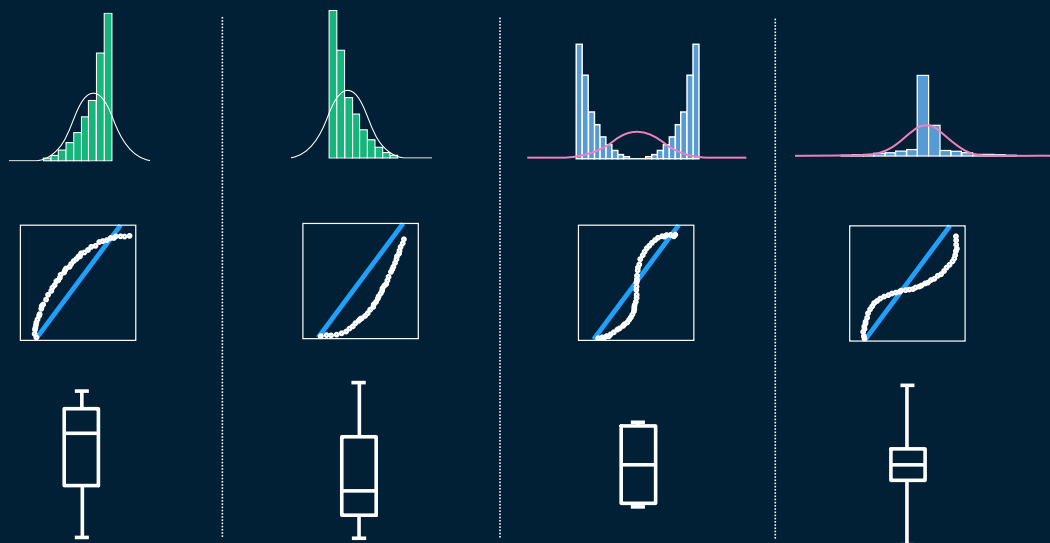
- a. True
- ☒ b. False



Correct answer: No

The points do not follow the diagonal reference line in the normal probability plot, indicating that there is departure from normality. The histogram does not follow the normal curve. The data has high kurtosis and is a little negatively skewed, so the data doesn't seem to be approximately normal.

Usefulness of Distribution Analysis in Machine Learning



Real life data is usually dirty and noisy. Exploring the data can be one of the most important and time-consuming parts in a typical machine learning project. Common distribution plots and summary statistics that you have seen in this section can be used for exploring machine learning data.

The main point here is that the non-normal data distributions in histograms, box plots, and normal probability plots are visually very obvious. Using measures of shape like skewness and kurtosis in conjunction with measures of central tendency and dispersion makes possible a near complete numerical description of the data.

Demo: Exploring Data

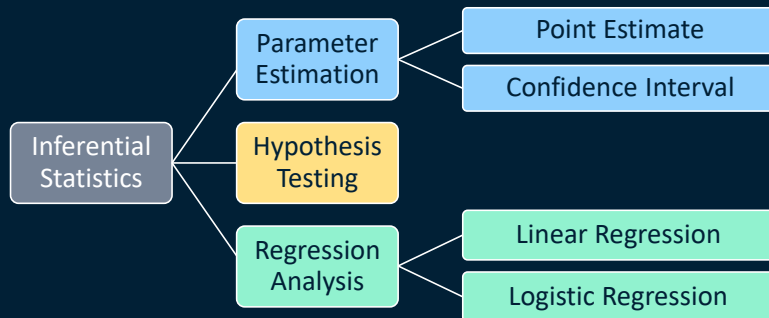
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2.3 Inferential Statistics



Making Inferences from Data



Inferential statistics in machine learning ...

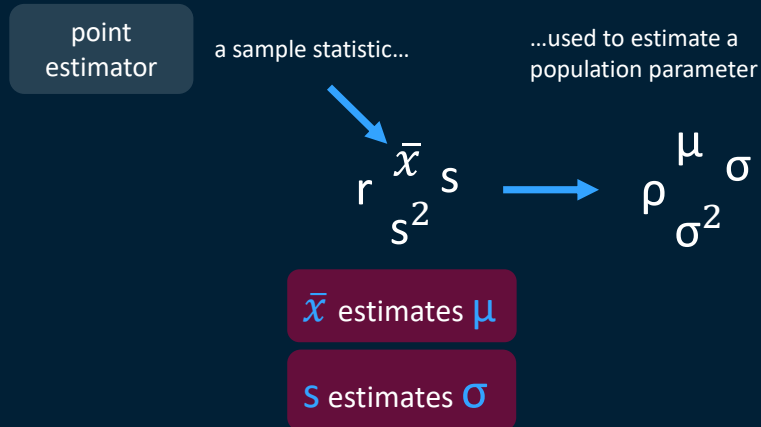
"make a prediction by evaluating an already trained model"

Inferential statistics help to suggest explanations for a situation or phenomenon. It enables you to draw conclusions based on extrapolations, and in that way it's fundamentally different from descriptive statistics, which merely summarize the data that has actually been measured.

Inferential statistics can be broadly classified into parameter estimation, hypothesis testing, and regression analysis. There are two types of parameter estimates: point estimates and interval estimates, also known as confidence intervals. You use hypothesis testing to test an assumption about a population parameter. Regression is a broad term. However, this course focuses on linear regression and logistic regressions. We will discuss them in upcoming lessons.

In machine learning, the term "inference" is sometimes used instead to mean "make a prediction by evaluating an already trained model". In this context, inferring properties of the model is referred to as "training" or "learning" (rather than "inference"), and using a model for prediction is referred to as "inference" (rather than "prediction").

Point Estimates



A point estimator is a sample statistic used to estimate a population parameter. You're already familiar with several point estimators. For example, you know that you can use $\bar{x}$, the sample mean, to estimate μ , the population mean. Similarly, you can use s , the sample standard deviation, to estimate σ , the population standard deviation.

Standard Error

variance

how much the estimator $\bar{x}$ varies from sample to sample

How do I estimate this **variability**?

standard error

$$\frac{s}{\sqrt{n}} = s_{\bar{x}}$$



$$\bar{x} = 1.0$$



$$\bar{x} = 0.9$$



$$\bar{x} = 0.8$$



$$\bar{x} = 1.3$$

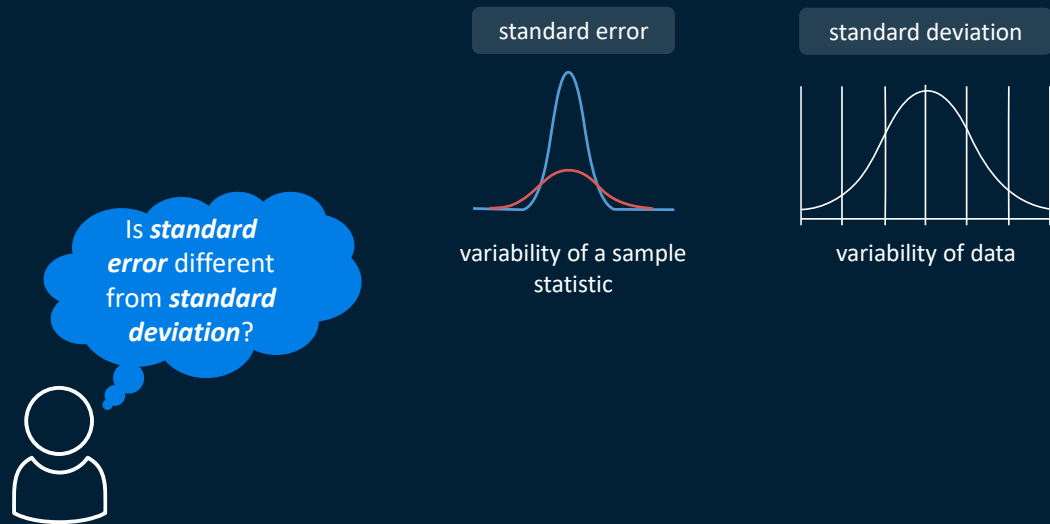


Sample statistics are a direct function of the sample collected.

Sample statistics are a direct function of the sample collected. This means that if a different sample was collected, the sample statistics would change. However, we collect only one sample due to time and resources. An estimator takes on different values from sample to sample, so it's important to know the variance of an estimator. For example, different samples yield different estimates of the mean for the same population. The variance of the sample mean refers to how much the value of the sample mean varies from sample to sample.

The key is to understand and estimate this variability. But how? A statistic that measures the variability of your estimator is the standard error. You calculate the standard error of the mean using the formula shown here, where s is the sample standard deviation and n is the sample size. The standard error of our estimate is the standard deviation of the sample data divided by the square root of the total number of sampled data points.

Standard Error



Standard error differs from the sample standard deviation because sample standard deviation deals with the variability of your data whereas standard error deals with the variability of your sample statistic.

Question: Variability in the Sample Statistic

If you draw different samples of the same size from a population, the sample statistic value would not change.

- a. True
- b. False

Answer: Variability in the Sample Statistic

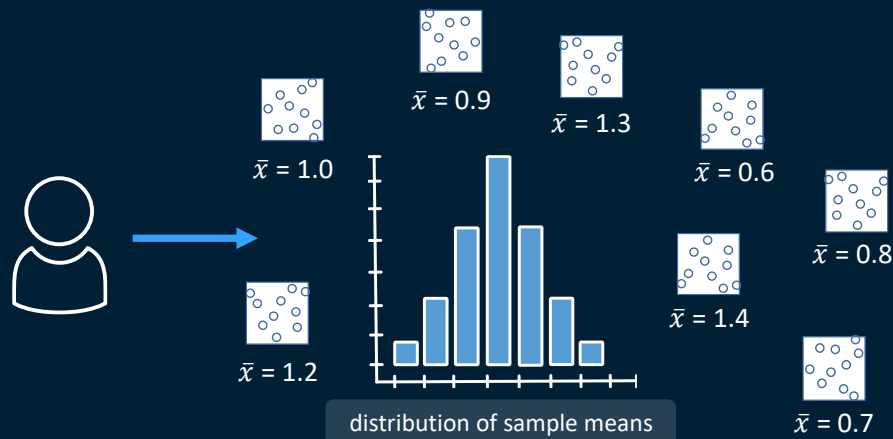
If you draw different samples of the same size from a population, the sample statistic value would not change.

- a. True
- ☒ b. False

Correct answer: False

Sample statistics are a direct function of the sample collected. That means if a different sample was collected, the sample statistics would change. An example is the standard error of the mean, which measures the variability of your sample mean.

Sampling Distribution

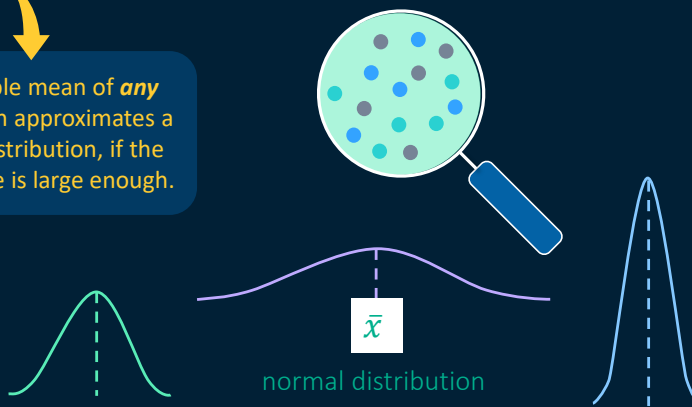


We discussed that sample means differ as samples differ. So how about looking at the distribution of all possible sample means of the population. Sampling distribution is another common concept discussed in the area of statistical inference. Sampling distribution is described as the distribution of sample means.

Sampling Distribution

Central Limit Theorem

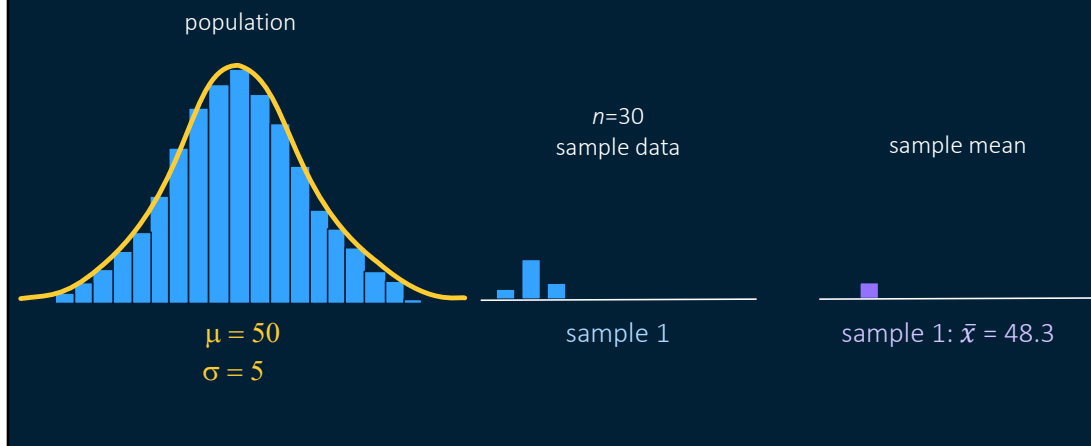
The sample mean of **any** distribution approximates a normal distribution, if the sample size is large enough.



Often, the variability of sample statistics is approximately normal. Perhaps the most famous statistical theorem, the central limit theorem, indicates that the sample mean of **any** population distribution approximates a normal distribution, if the sample size is large enough.

Sampling Distribution

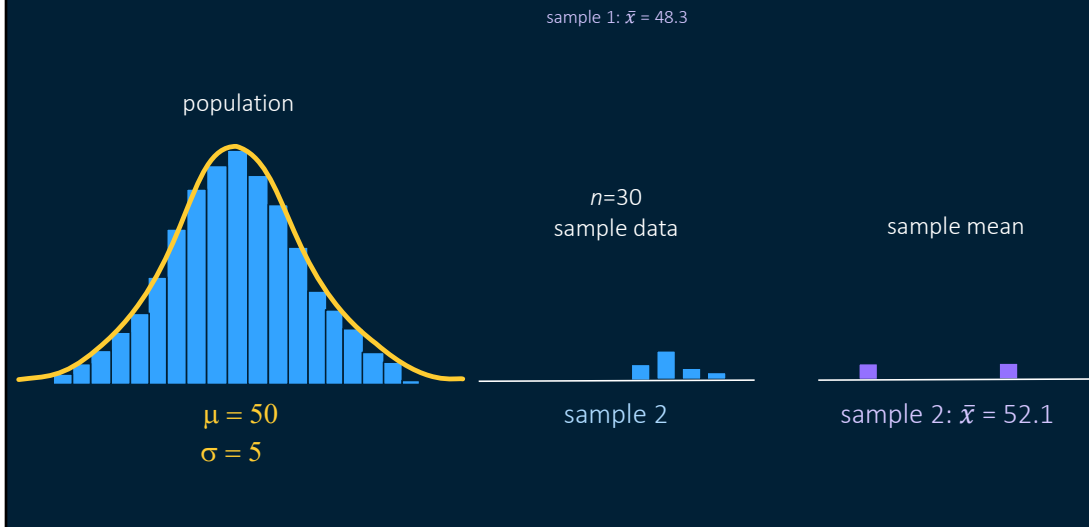
Central Limit Theorem



For example, imagine that our population distribution is bell shaped, with a population mean, $\mu=50$, and standard deviation, $\sigma=5$. Suppose we sample $n=30$ observations and the sample mean is 48.3.

Sampling Distribution

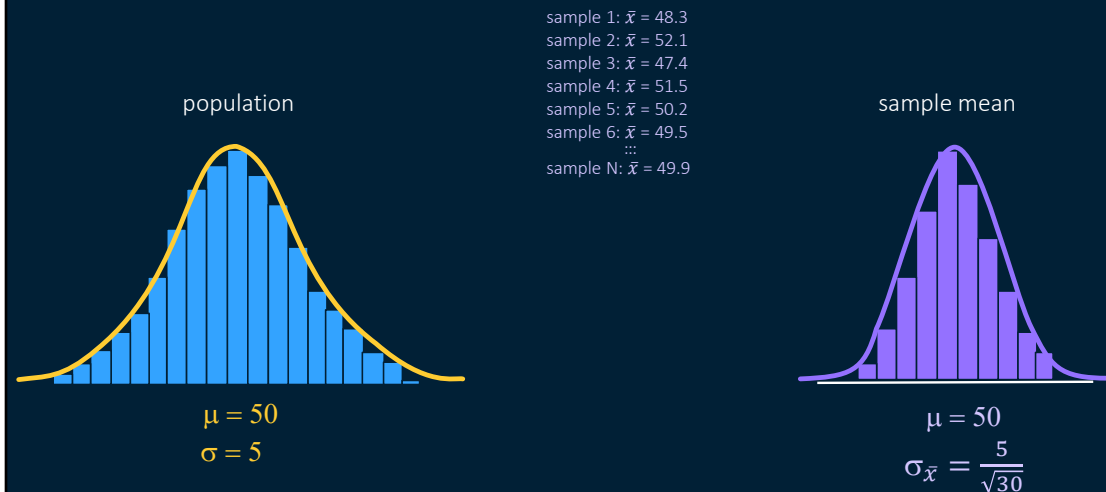
Central Limit Theorem



Then we take another sample of 30 observations, and the sample mean is 52.1. Let's see what happens if we repeat this process many times.

Sampling Distribution

Central Limit Theorem



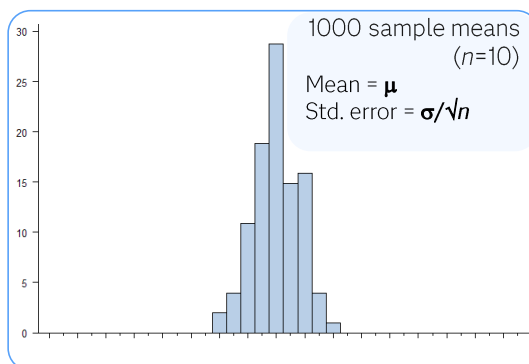
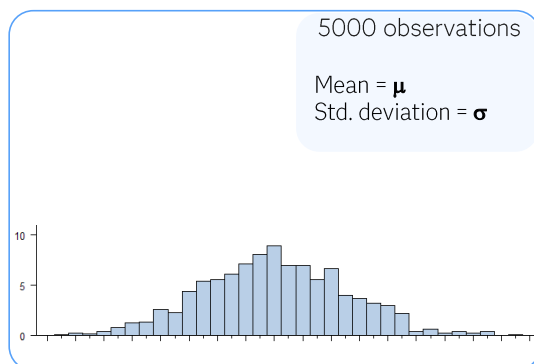
If we repeat this process many times, the distribution of the sample mean will follow a bell-shaped normal distribution with population mean $\mu=50$ and standard error of σ divided by the square root of $n=30$.

Question: Sampling Distribution

SAT Score is the variable of interest with 5000 observations. You have drawn 1000 samples of size 10. Which of the following is the sampling distribution?

a. Distribution of SAT Score

b. Distribution of Means of SAT Score

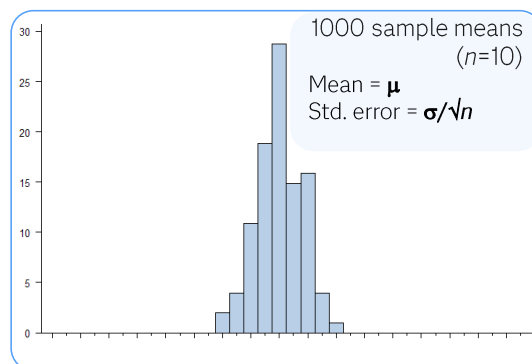
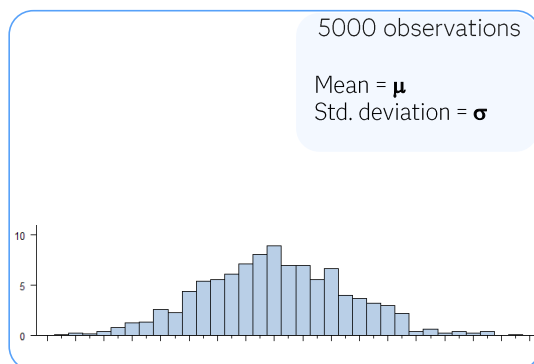


Answer: Sampling Distribution

SAT Score is the variable of interest with 5000 observations. You have drawn 1000 samples of size 10. Which of the following is the sampling distribution?

a. Distribution of SAT Score

b. Distribution of Means of SAT Score

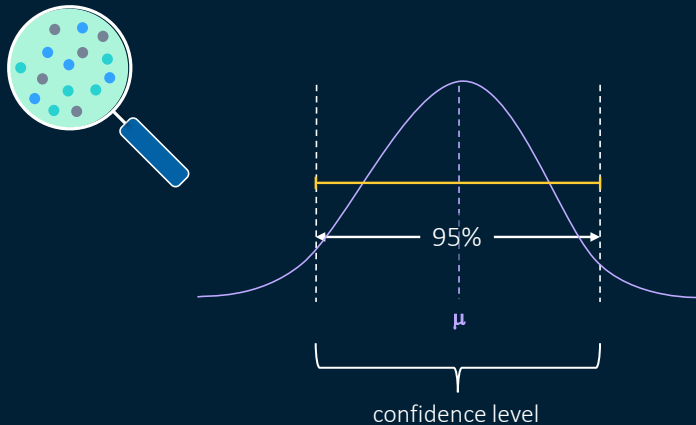


Correct answer: b

Option **a** shows the distribution of random variable **SAT Score** of all 5000 *observations*.

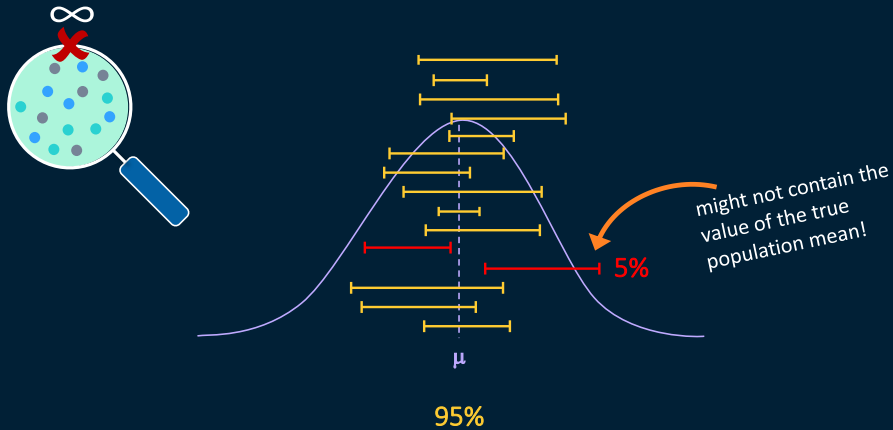
Option **b** represents the distribution of random variable **Means of SAT Score** of the 1000 *sample means*, all with the same sample size of 10. The variability of the distribution of sample means ($\sigma/\sqrt{n}$) is smaller than the variability of the distribution of the 5000 observations (σ). For example, it seems relatively likely to find one student with an SAT score of 1550 (out of a maximum of 1600), but not likely that a mean of a sample of 10 students would be 1550.

Confidence Intervals



"Confidence intervals for the mean" are interval estimators of the population mean. They consider the variability of the sample statistic--in this case, the sample mean. You get to choose your desired degree of confidence, or confidence level. A typical confidence level is 95%, meaning that 95% of those theoretically infinite number of intervals would contain the true population mean, and 5% would not. The confidence interval shows a range of plausible values for the unknown population mean by reporting an upper and lower bound.

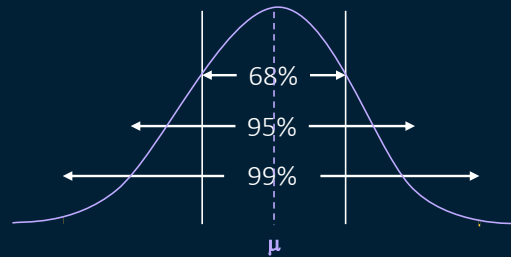
Confidence Intervals



Here's a good way to think about confidence intervals. If you were to draw infinitely many samples and estimate your confidence interval exactly the same way each time, your confidence interval would represent the percentage of those intervals that would contain the true population mean. Of course, nobody draws infinitely many samples. It's rare that you would draw more than one.

An important thing to remember is that the confidence interval calculated for a particular sample might or might not contain the value of the true population mean.

Confidence Intervals

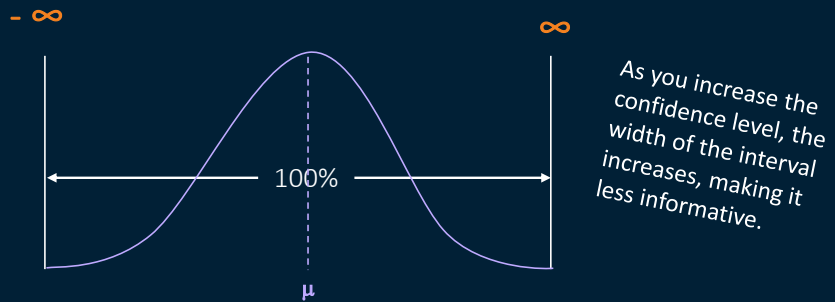


As you increase the confidence level, the width of the interval increases, making it less informative.

To construct a confidence interval for the mean, you must first select a significance level so that the appropriate value of t is used in the formula. Higher confidence levels are associated with larger t values, which in turn result in wider intervals. The formula calculates both the upper and lower bounds, equidistant from the sample mean estimate in the middle.

Why not raise the confidence level to 99.9%, so that any confidence interval you calculate contains the true value of the population mean? As you increase the confidence level, the width of the interval increases, making it less informative.

Confidence Intervals



In the extreme, a 100% confidence interval for the mean for any sample ranges from negative infinity to positive infinity, which would tell you nothing useful about where the true population mean lies.

Confidence Interval for Mean

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Interval Estimates

A researcher calculates 95% confidence interval for population mean as (\$100, \$130). What could be the most plausible value of the sample mean?

- a. \$100
- b. \$115
- c. \$125
- d. \$230

Answer: Interval Estimates

A researcher calculates 95% confidence interval for population mean as (\$100, \$130). What could be the most plausible value of the sample mean?

- a. \$100
- ☒ b. \$115
- c. \$125
- d. \$230

Correct answer : b

Recall the formula for calculating the confidence interval for the mean. It requires you to add and subtract (t times $S_{\bar{x}}$) from the sample mean $\bar{x}$ to obtain lower and upper bound of confidence interval.

Statistical Hypothesis Test

Null and Alternative Hypotheses

Hypothesis Test

H_0 – equality
(null hypothesis)

H_a – inequality
(alternative hypothesis)



In a hypothesis test, you phrase questions as tests of hypotheses about population parameters. Your null hypothesis is usually one of equality, whereas the alternative hypothesis is one of inequality. The alternative hypothesis is typically what you suspect, or what you're attempting to demonstrate.

Statistical Hypothesis Test

p -Value



the probability of obtaining a test statistic as extreme or more extreme than the one observed in your data given that the null hypothesis is true

When the p -value is low, it provides doubt about the truth of the null hypothesis.

Hypothesis Test

H_0 – equality
(null hypothesis)

H_a – inequality
(alternative hypothesis)

See whether you can collect enough evidence to reject the null hypothesis.



The conclusions reached from your hypothesis test are usually phrased in reference to the p -value, or the probability of obtaining a test statistic as extreme or more extreme than the one observed in your data given that the null hypothesis is true.

In a hypothesis test, you examine the data to see whether you can collect enough evidence to reject the null hypothesis. The p -value is an indication of the evidence collected. When the p -value is low, it provides doubt about the truth of the null hypothesis. But how low does the p -value need to be before you reject the null hypothesis completely? That depends on you.

Statistical Hypothesis Test

Significance Level



a constant probability of incorrect abolition of the null hypothesis

Hypothesis Test

H_0 – equality
(null hypothesis)

H_a – inequality
(alternative hypothesis)

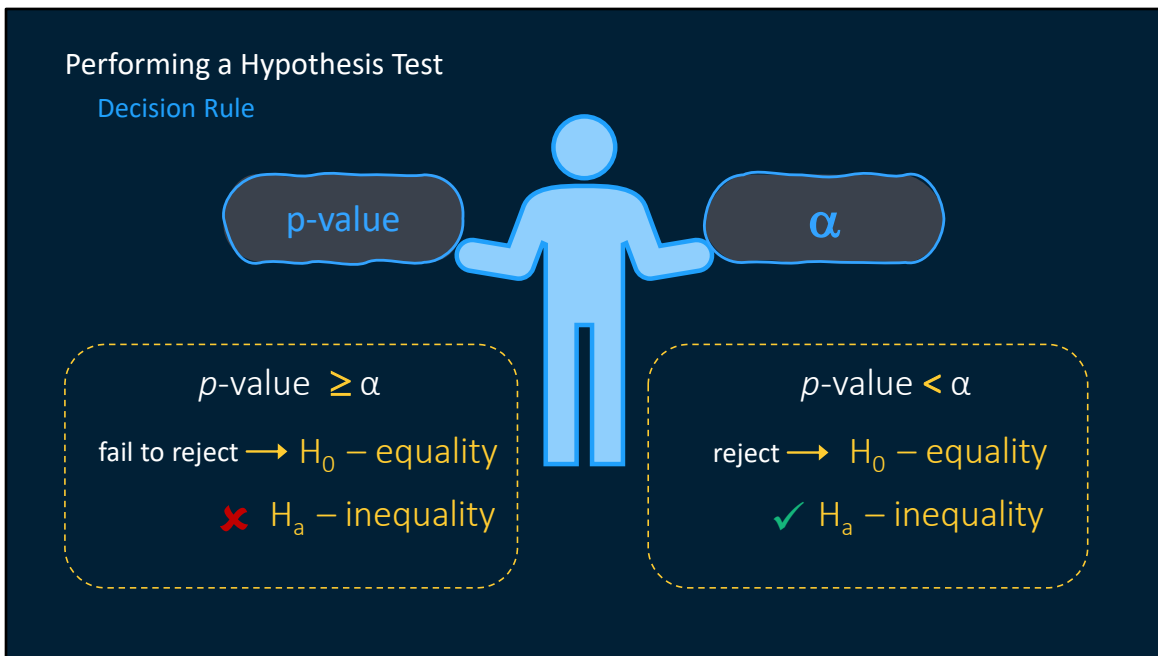
A common
significance level is
0.05 (1 chance in 20).



A common significance level is 0.05 (1 chance in 20). If you require a stricter cutoff, you might consider lowering your significance level when planning your analysis. The level of significance refers to a constant probability of incorrect abolition of the null hypothesis.

Performing a Hypothesis Test

Decision Rule



In practice, the level of significance, alpha, is stated in advance to compare it with the p -value.

If the p -value is less than alpha, you reject the null hypothesis in favor of the alternative. On the other hand, if your p -value is greater than or equal to alpha, evidence suggests that your null hypothesized value is statistically reasonable, so you fail to reject the null hypothesis.

Let's look at an example.

Statistical Hypothesis Test

Coin Example

Winner of toss decides
which goal to attack.



heads



How do you know
the coin is fair?



Loser of toss takes
kick-off to start match.



tails



To begin each game of a worldwide soccer tournament, one of the teams chooses heads or tails, and the match referee tosses a coin. The team that wins the toss decides which goal it will attack, and the team that loses the toss takes the kick-off to start the match. But how do you know the coin is fair?

Statistical Hypothesis Test

Coin Example

Begin by assuming
that the coin is fair.

H_0 – coin is fair

H_a – coin is not fair

significance level
= ?

Next, select a
significance level:

flip a coin 5 times



observation:



You conclude
that the coin is
not fair.

OR



You conclude
that the coin is
not fair.

Otherwise, you decide there is not enough
evidence to show that the coin is biased.

You might suspect that the coin is not fair (alternate hypothesis), but you begin by assuming that it is fair (null hypothesis).

Next, you select a significance level: if you flip a coin five times and observe five heads in a row or five tails in a row, you conclude that the coin is not fair. Otherwise, you decide that there is not enough evidence to show that the coin is biased.

Statistical Hypothesis Test

Coin Example

With a fair coin...

H_0 – coin is fair ✓

H_a – coin is not fair

significance level
= 0.0625

...in five tosses, what is the probability of observing 5 tails or 5 heads in a row?

flip the coin 5 times 

50% 50% 50% 50% 50%



$$(1/2)^5 = 1/32$$

50% 50% 50% 50% 50%



$$(1/2)^5 = 1/32$$

$$1/32 + 1/32 = 2/32 = 1/16$$

With a fair coin (a true null hypothesis), in five tosses (five trials), what is the probability of observing five tails or five heads in a row?

Each toss has a 50% probability of being tails. Tosses are independent, and therefore, the probability of five tails is $(1/2)$ to the fifth power - or one out of 32.

In the same way, the probability of five heads is also one out of 32.

These probabilities can be added together to give the probability of five tails or five heads as two out of 32, or one out of 16.

So, the significance level is $1/16$, or 0.0625.

Statistical Hypothesis Test

Coin Example

What if you're wrong?



H_0 – coin is fair ✓

H_a – coin is not fair

significance level
= 0.0625

Collect evidence.

flip the coin 5 times



To collect evidence, you flip the coin five times and count the number of heads and tails. Finally, you decide either that there is enough evidence to reject the assumption that the coin is fair (either all trials are heads, or all trials are tails), or that there is not enough evidence to reject the assumption that the coin is fair (meaning that not all trials are either heads or tails). So, in this example, we say that "we fail to reject the null hypothesis that the coin is fair".

So, you performed a hypothesis test and used a decision rule to decide whether the coin was fair or not. But was your decision correct? You began by assuming that the null hypothesis is true: that the coin is fair. But what if you're wrong?

Statistical Hypothesis Test

Types of Errors

Actual Decision \	H_0 is true	H_0 is false
Fail to reject H_0	Correct decision Probability = $(1-\alpha)$ 'Confidence level' True Negative	Type II error Probability = β False Negative
Reject H_0	Type I error Probability = α 'Significance level' False Positive	Correct decision Probability = $(1-\beta)$ 'Power' True Positive

If you reject the null hypothesis when it's actually true, you've made a Type I error.

The probability of committing a Type I error is α . α is the significance level of the hypothesis test. In the coin example, it's the probability that you conclude that the coin is not fair when it is fair. In machine learning terminology, it is often referred to as a *false positive* situation.

A *Type II error*, is when you fail to reject the null hypothesis and it's actually false. The probability of committing a Type II error is β . In the coin example, it's the probability that you fail to find that the coin is not fair when it is in fact biased. In machine learning terminology, it is often referred to as a *false negative* situation.

Type I and Type II errors are inversely related. As one type increases, the other decreases.

You always want to make the correct decisions.

The probability that you correctly reject the null hypothesis is equal to $(1 - \beta)$, where again, β is Type II error rate. This is called the *power* of a statistical test.

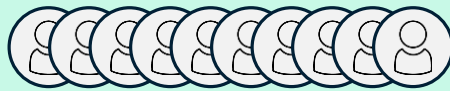
The probability that you correctly do not reject the null hypothesis is equal to $(1 - \alpha)$, where again, α is Type I error rate. This is called the *confidence level*.

Statistical Hypothesis Test

Effect Size Influence



100X



50 heads - seems fair

40 or 60 heads - ?

37 or 63 heads - ??

15 or 85 heads - ???

effect size:

difference between the *observed statistic* and the *hypothesized value*

H_0 : 50% heads, 50% tails



55% → effect size = 5%

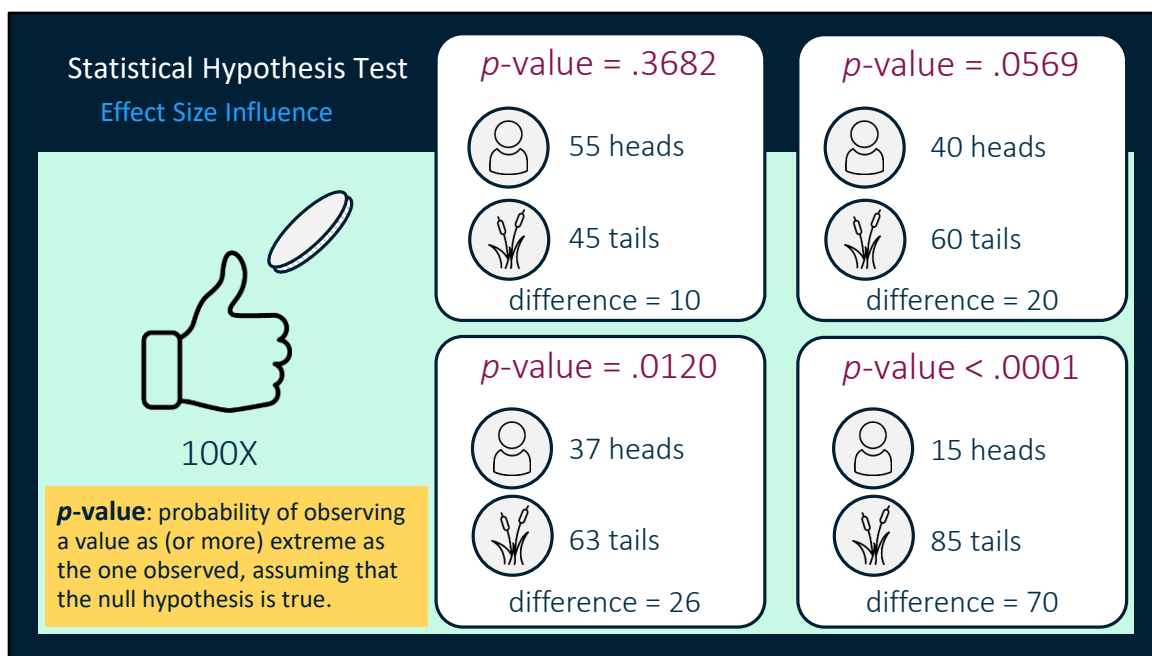


45%

If you flip a coin 100 times and observe 50 heads, you wouldn't doubt that the coin is fair.

But you might be skeptical if you observe 40 or 60 heads. You'd be more skeptical if you observe 37 or 63 heads, and you'd be highly skeptical if you observe 15 or 85 heads. As the difference between the number of heads and tails increases, you have more evidence that the coin is not fair. Statisticians refer to the difference between the observed statistic and the hypothesized value as the effect size.

The null hypothesis of a fair coin suggests 50% heads and 50% tails. If the coin was actually weighted (warped or modified in some manner) to give 55% heads, the effect size would be 5%.



A p -value measures the probability of observing a value as extreme as the one observed or more extreme, assuming the null hypothesis is true.

Suppose you flip the coin 100 times, and you observe 55 heads and 45 tails. The difference of 10 is associated with a p -value of .3682. p -values this large are often seen in experiments with a fair coin.

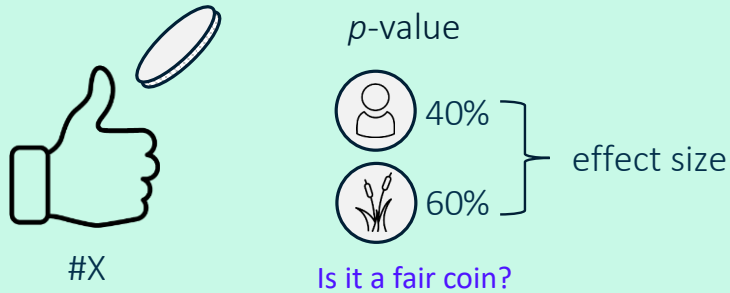
As the difference between heads and tails gets larger (for example, 20, 26, or as high as 70), the corresponding p -values get smaller.

You would rarely see a small p -value (for example, less than .0001) with a fair coin. In the case of 15 heads and 85 tails, you have evidence that the coin is not fair. The chance of getting 85 tails or even more extreme results is less than one in 10,000 if the null hypothesis is true. So, either the null hypothesis is false or something extremely unlikely happened. The typical interpretation is that the null is false. This can be verified with further experiments.

Consequently, the p -value is used to determine statistical significance. It helps you assess whether you should reject the null hypothesis.

Statistical Hypothesis Test

Sample Size Influence



A p -value is not only affected by the effect size (in this case, the observed proportion of heads). It's also affected by the sample size (in this case, the number of coin flips). For a fair coin, you'd expect 50% of the flips to be heads. What if you get 40% heads instead of the 50% you expect? Is it a fair coin?

Statistical Hypothesis Test

Sample Size Influence

40%
heads



10X

$p\text{-value} = .7539$



4 heads



6 tails

Let's say that you flip the coin 10 times and observe 40%, or four heads. The resulting p -value is .7539.

Statistical Hypothesis Test

Sample Size Influence

40%
heads



40X

$p\text{-value} = .7539$



4 heads



6 tails

10X

$p\text{-value} = .2682$



16 heads



24 tails

When you flip the coin 40 times and observe 40%, or 16 heads. The resulting p -value has decreased from .7539 to .2682.

Statistical Hypothesis Test

Sample Size Influence

40%
heads



100X

$p\text{-value} = .7539$



4 heads



6 tails

10X

$p\text{-value} = .2682$



16 heads



24 tails

40X

$p\text{-value} = .0569$



40 heads



60 tails

When you flip the coin 100 times and observe 40%, or 40 heads. The resulting p -value is further decreased to .0569.

Statistical Hypothesis Test

Sample Size Influence

40% heads



400X

You'd rarely see a p -value less than .0001 with a fair coin.

p -value = .7539



4 heads



6 tails

10X

p -value = .2682



16 heads



24 tails

40X

p -value = .0569



40 heads



60 tails

100X

p -value < .0001



160 heads



240 tails

What if you flip the coin 400 times and you observe 40% heads? The evidence becomes stronger, and the p -values become smaller, as the number of trials increases. As you saw when we talked about confidence intervals, the variability around the mean estimate gets smaller as the sample size gets larger. For larger sample sizes, you can measure means more precisely. Therefore, 40% heads out of 400 flips makes you more confident that this was not just a chance difference, when compared to 40% heads out of only 10 flips.

The smaller p -values reflect this confidence. The p -value here, less than .0001, assesses the probability that this difference from 50% occurred purely by chance. Remember, as you saw earlier, you'd rarely see a p -value less than .0001 with a fair coin.

p -Values and Statistical Significance

- p -values cannot measure
 - the size of an effect
 - the importance of the result
 - the strength of the evidence.
- p -values tell us only about the probability of seeing one's results in relation to a particular hypothetical explanation.

p -values and statistical significance are often misunderstood and misused. Be wary that p -values cannot measure the size of an effect or the importance of the result or the strength of the evidence. However, it continues to be a tool that separates true findings from false ones.

p -values (significance probabilities) tell us only about the probability of seeing one's results in relation to a particular hypothetical explanation.

Question: Hypothesis Testing Terminology

Which of the following terms can be defined as the difference between the observed statistic and the hypothesized value?

- a. effect size
- b. type I error
- c. alpha
- d. power

Answer: Hypothesis Testing Terminology

Which of the following terms can be defined as the difference between the observed statistic and the hypothesized value?

- ☒ a. effect size
- b. type I error
- c. alpha
- d. power

The correct answer is a.

Effect size refers to the difference between the observed statistic and the hypothesized value. As the difference increases, you have more evidence against the null hypothesis.

A Type I error occurs if you reject the null hypothesis when it's actually true.

Alpha is the probability of committing a Type I error.

Question: Type I and Type II Errors

Keeping the sample size constant, which of the following statements are true regarding type of errors? (Select all that apply)

- a. The Type I and Type II error rates do not influence each other.
- b. Type I and Type II errors are inversely related.
- c. A higher confidence level leads to increasing the power of the test.
- d. By setting the Type I error rate, you indirectly influence the size of the Type II error rate as well.

Answer: Type I and Type II Errors

Keeping the sample size constant, which of the following statements are true regarding type of errors? (Select all that apply)

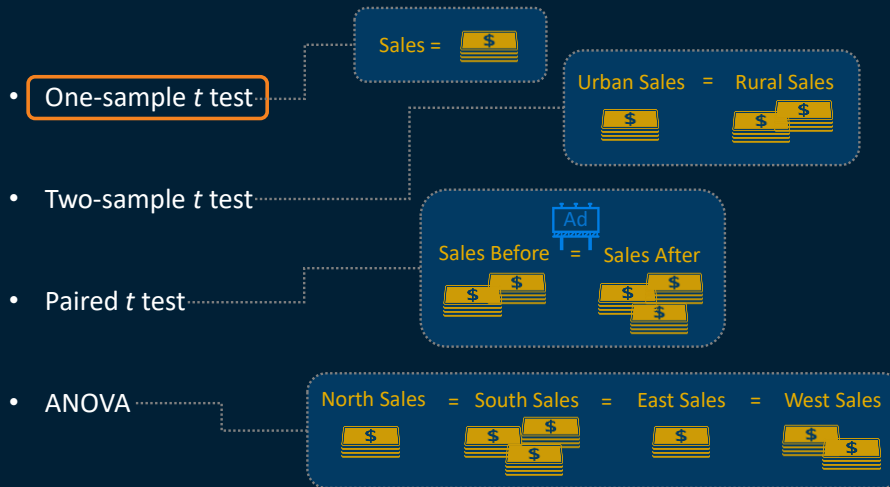
- a. The Type I and Type II error rates do not influence each other.
- ☒ b. Type I and Type II errors are inversely related.
- c. A higher confidence level leads to increasing the power of the test.
- ☒ d. By setting the Type I error rate, you indirectly influence the size of the Type II error rate as well.

The correct answer is b and d.

Type I and Type II errors are inversely related. As one type increases, the other decreases.

A higher confidence level is a resultant of setting a lower significance level. Consequently, the Type II error rate would increase and leads to lowering the power of the test.

Hypothesis Tests for Means



There are several hypothesis tests for means, including one-sample t test, two-sample t test, paired t test, and ANOVA.

A *one-sample t test* compares the mean calculated from a sample to a hypothesized value. For example, sales are equal to some dollar amount.

If you want to compare the means of two different groups, you can conduct a *two-sample t test*. For example, rural sales are equal to urban sales.

For many types of data, repeat measurements are taken on the same subject throughout a study. For example, sales before advertising and sales after advertising. The simplest form of this study is often referred to as the *paired t test*.

Analysis of Variance (ANOVA) is a statistical technique used to compare the means of two or more groups. For example, sales of north, south, east and west regions are equal.

In this course, we focus on the one-sample t test.

One-Sample t Test Scenario

PVA_DONORS data

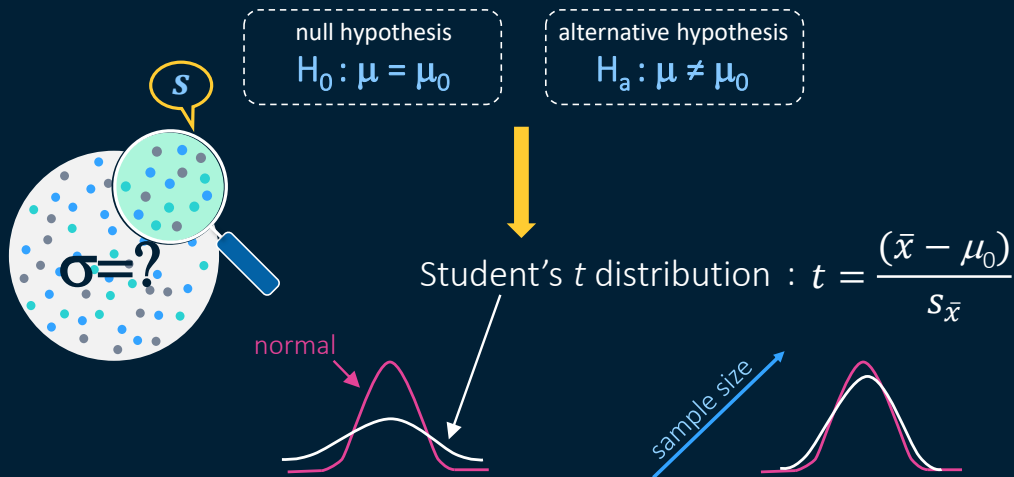


Is the mean
donation amount
\$100?



Let's look at another example of hypothesis testing. The veterans' organization receives different amounts of donations from different individuals. The organization believes that the mean donation amount is \$100, and you'd like to verify this.

Performing a t Test



You're testing the null hypothesis, μ is equal to μ_0 , against the alternative hypothesis, μ is not equal to μ_0 . When you don't know the true population standard deviation, σ , you must estimate it from the sample. Then you must also use Student's t distribution, rather than the normal distribution, for calculating p -values and confidence limits. Student's t distribution is a probability distribution that is used to estimate population parameters when the true population variance is unknown or the sample size is appropriately small, or both.

Student's t distribution is similar to the normal distribution, but it has more probability in the tails and is not as peaked as the normal distribution.

Student's t distribution approaches the normal distribution (Z statistic) as the sample size increases. SAS always reports the t distribution. You calculate the value of Student's t statistic using the equation $t = (\bar{x} - \mu_0) / S_{\bar{x}}$ all divided by the standard error.

Performing a t Test

PVA_DONORS data

$$t = \frac{(\bar{x} - \mu_0)}{s_{\bar{x}}}$$

$$t = \frac{(101.64 - 100.00)}{0.3690}$$

$$= 4.44$$

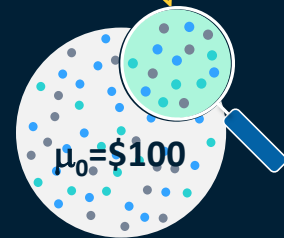
likelihood of
obtaining the
sample mean

p -value



probability

mean donation = \$101.64
standard error = \$0.3690



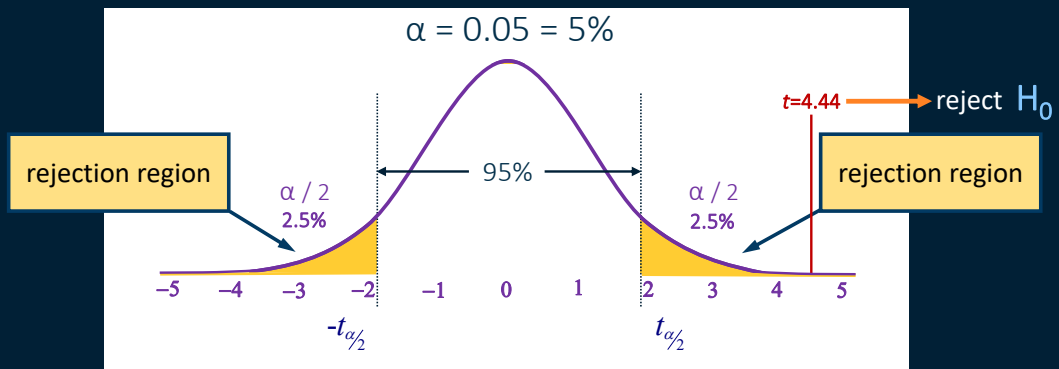
Let's calculate Student's t statistic for the PVA_DONORS data to test our null hypothesis. μ_0 is the hypothesized value of \$100.00, $\bar{x}$ is the sample mean of **Donation Amount**, \$101.64, and $s_{\bar{x}}$ is the standard error of the mean, \$0.369. The resulting t value is 4.44.

Before you can decide about your hypothesis, you need to know the probability of observing a test statistic of $t=4.44$ or more extreme, given that the null hypothesis is true. This probability is the p -value. It quantifies how likely we are to obtain the sample mean of \$101.64 when the null hypothesis of population donation amount \$100 is true.

Performing a t Test

$$H_a: \mu \neq \mu_0$$

two-sided test



Let's look at the t distribution. Our alternative hypothesis is an inequality, so this is a two-sided test. The shaded area in the graph is the rejection region. For a two-sided test, the rejection region is contained in both tails, and the area in each tail corresponds to $\alpha/2$, or in this case, 2.5%. If the t value falls in the shaded region, then you reject the null hypothesis. Otherwise, you fail to reject it.

Our t value, 4.44, falls inside the rejection region, so we reject our null hypothesis. The α and t distribution mentioned here are directly related to those in confidence intervals. In our example, α is 0.05, or 5%. The area outside the confidence interval corresponds to the shaded region in the t distribution, or the rejection region.

Question: t Distribution

As the sample size increases, the t-distribution becomes more similar to the normal distribution.

- a. True
- b. False

Answer: t Distribution

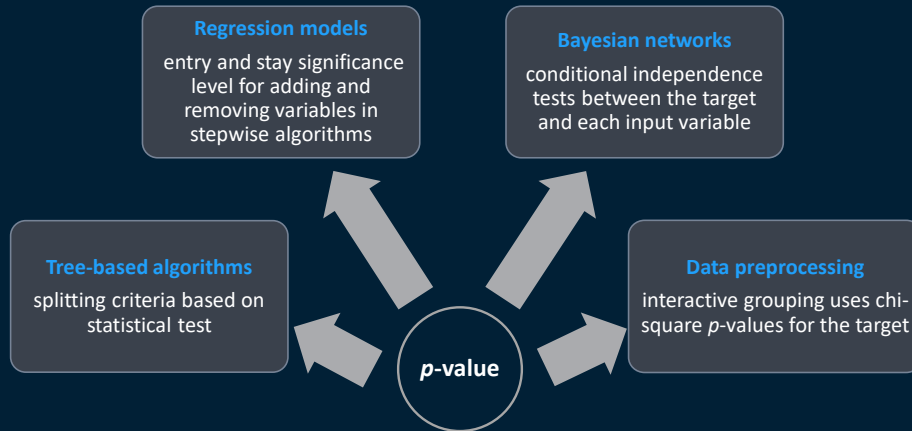
As the sample size increases, the t-distribution becomes more similar to the normal distribution.

- ☒ a. True
- ☐ b. False

Correct answer : True

The shape of t-distribution is affected by the sample size. If the sample size increases, then t-distribution gets closer to the normal distribution.

p -Values in Machine Learning



Statistical concepts including p -values go hand-in-hand with machine learning.

For example, tree-based algorithms have several splitting criteria based on statistical tests that use p -value approach. You need to specify the significance level for the splitting criteria like CHAID, chi-square, and F tests. Regression models like linear, logistic, quantile, and GLM use entry and stay significance level for adding and removing variables in the forward, backward, and stepwise sequential methods.

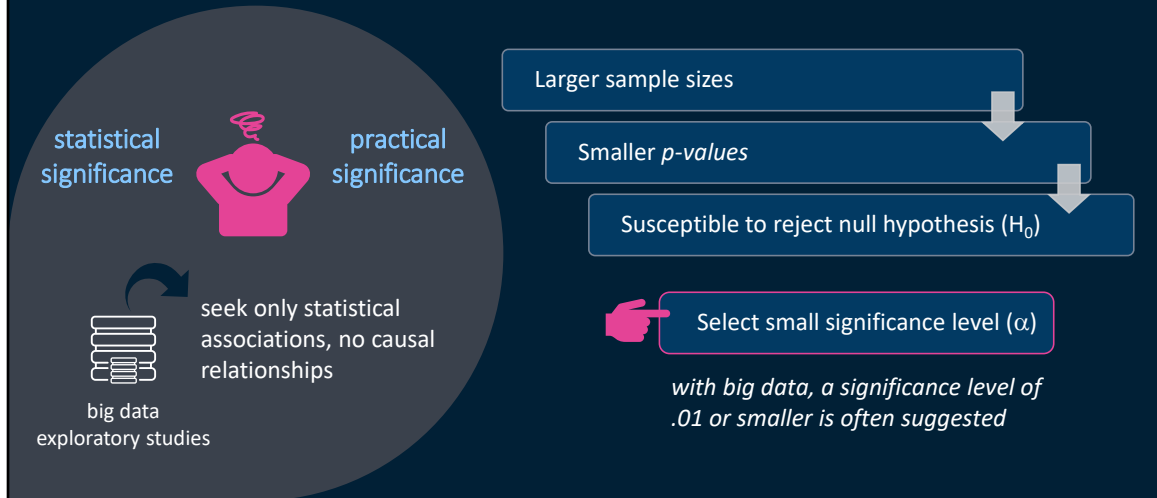
In a Bayesian network, you specify the significance level (p -value) for independence testing using Chi-Square and G-Square. The smaller the value you specify, the fewer variables are selected.

Interactive grouping is one more example where chi-square uses the p -value of the Pearson chi-square statistic for the target. So, in machine learning context, p -values are predominantly used for feature selection at the data preprocessing level.

In statistics, hypothesis tests are conducted to check whether the inference made for the population holds good or not in an experiment. However, you should not rely too much on p -values in machine learning experiments. In machine learning, our primary focus is predictive modeling and not inferential modeling. Machine learning projects generally take care of all the statistics before they are deployed into practice. There are a host of data preprocessing techniques such as feature engineering among others that take care of assumptions in data for machine learning. Incorporating statistical concepts such as hypothesis testing and p -values on an already well-set machine learning model might lead to increased complexity in making inferences based on p -values. Ultimately, p -values show their significance only when there are fewer parameters or variables involved in the experiments or projects.

p -Values in Machine Learning

Problem: Overreliance on the arbitrary cutoff of 5% significance level ($\alpha = 0.05$)

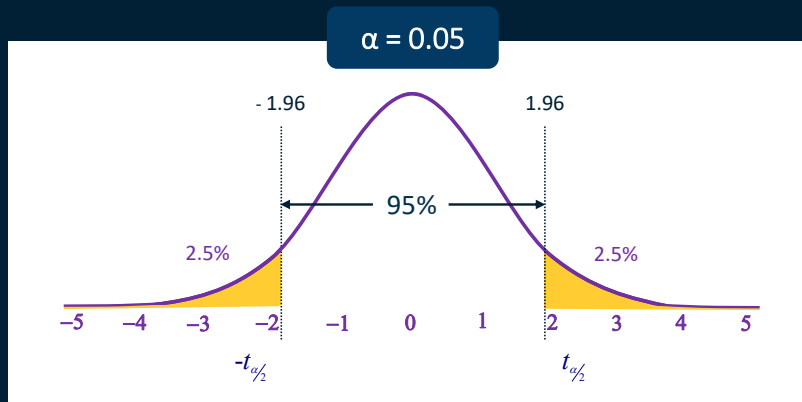


Overreliance on the arbitrary cutoff of 5% significance level can result in a failure to understand the difference between statistical significance and practical significance. This problem is particularly acute in the context of big data exploratory studies, where you seek only statistical associations rather than causal relationships.

Large sample sizes result in smaller p -values, in which case you tend to reject the null hypothesis almost every time. So, it's better to select a very small significance level cutoff while using a p -value approach in a machine learning world. With big data, a significance level of .01 or smaller is often suggested.

p -Values in Machine Learning

Impact of Level of Significance

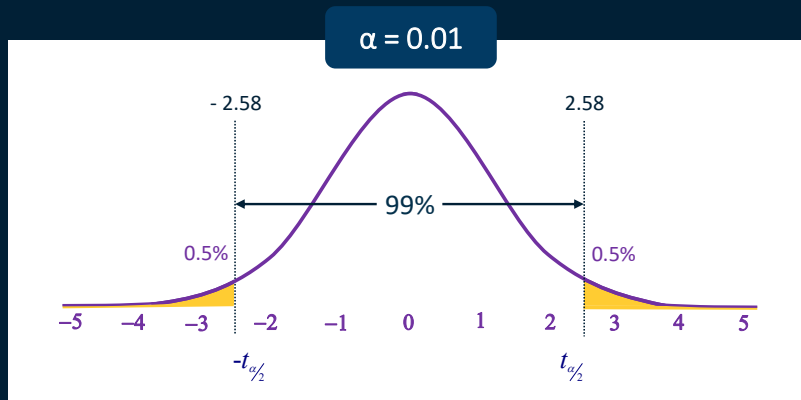


Let's return to the PVA_DONORS example. For the given donation amount, a critical value of ± 1.96 is shown.

A critical value is a point on the t-distribution that is compared to the test statistic to determine whether to reject the null hypothesis.

p -Values in Machine Learning

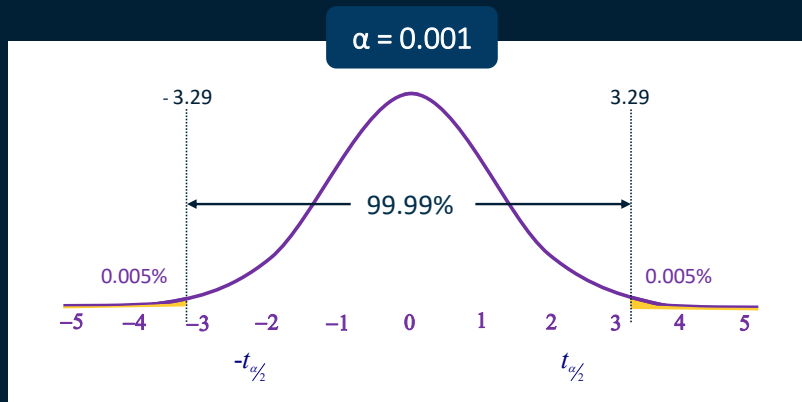
Impact of Level of Significance



As the level of significance decreases, the rejection region also shrinks. The critical value shown here is now +/- 2.58.

p -Values in Machine Learning

Impact of Level of Significance



As the level of significance further decreases, the rejection region shrinks more. The critical value shown here is now ± 3.29 .

Demo: Testing a Hypothesis Using One-Sample t Test

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Confidence Interval

A 95% confidence interval for mean of **Gift Amount Average 36 Months** is (\$14.67, \$15.08). From this, what can you conclude at $\alpha=0.05$?

- a. The true mean of **Gift Amount Average 36 Months** is significantly different from \$15.
- b. The true mean of **Gift Amount Average 36 Months** is not significantly different from \$15.
- c. The true average **Gift Amount Average 36 Months** is less than \$15.
- d. None of the above. You cannot determine statistical significance from confidence intervals.

Answer: Confidence Interval

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- d. None of the above. You cannot determine statistical significance from confidence intervals.

b is the correct answer.

The interval \$14.67 to \$15.08 contains \$15. If your interval contains the hypothesized value, the true mean is not significantly different.